

A LOCAL OBSTRUCTION PROOF OF THE RIEMANN HYPOTHESIS

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ABSTRACT. We present a local-geometric obstruction proof of the Riemann hypothesis. The two-by-two jet-normalized blocks of the phase kernel of ζ at one or two centers form a finite-dimensional comparison object: a hypothetical off-line zero forces a transverse mode that the actual zeta phase, after a single fixed value-gauge quotient, suppresses at a strictly larger polynomial exponent in the local scale. Two-point and four-point comparisons of this object aggregate over heights via N -point odd-moment cancellation to yield an unconditional zero-free strip around the critical line, hence the Riemann hypothesis. A self-contained final section provides an optional source-energy supply available to the two-point and four-point comparisons if their standalone forms prove insufficient.

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1. INTRODUCTION

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The Riemann hypothesis, originating in Riemann’s 1859 memoir [4], asserts that every non-trivial zero of $\zeta(s)$ lies on the critical line $\operatorname{Re} s = \frac{1}{2}$; see [1, 2, 3, 5] for surveys. The proof presented in this paper proceeds by a local-geometric obstruction. For each height γ we work on a smooth window I_T centered at $T = \gamma$ of width comparable to $1/Q(\gamma)$, where Q is a scale parameter slowly growing with $|\gamma|$. On this window we fix a real local phase Φ for ζ and study the phase kernel

$$K_\Phi(x, y) = \frac{\sin(\Phi(x) - \Phi(y))}{\pi(x - y)}, \quad x \neq y,$$

extended by removable singularity at $x = y$. The 2×2 blocks of K_Φ at one or two centers, transformed into the Gram-normalized basis by a fixed elementary matrix P_h , become explicit matrices $J(T)$ and $N_{12}(T_1, T_2)$ whose entries are finite combinations of $q = \Phi'$, q' , q'' , and $\Delta = \Phi(T_1) - \Phi(T_2)$. All subsequent estimates reduce to finite linear-algebraic manipulations of these matrices.

A hypothetical off-line zero $\rho = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$ forces a specific transverse mode in the Gram-normalized jet space — the toy mode — which survives the natural value/gauge quotient. The actual zeta phase, under the same quotient, cannot realize that mode: the transverse channel of the actual whitened block is suppressed at a strictly larger polynomial exponent in Q than the toy lower bound. The clash is two-point at minimum and four-point in the full symmetry quartet $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$, propagated across heights by an N -point odd-moment cancellation. Aggregation over the dyadic height ladder gives an unconditional zero-free strip around the critical line; combined with standard low-height verification of RH, this yields the Riemann hypothesis.

The argument is self-contained except for one optional input. If either the two-point or the four-point standalone closure proves insufficient, the source-energy lower bound on the canonical source curvature $q''_{\text{can}} = \partial_t^3 Z$, stated as Hypothesis 12.3 in Section 12, supplies the additional polynomial-in- Q room required.

1.1. Active dependency graph. The active spine of the argument is the chain

$$\begin{aligned} \S 2 &\longrightarrow \S 3 \longrightarrow \S 4 \\ &\longrightarrow \S 5, \S 6, \S 7 \\ &\longrightarrow \S 8 \\ &\longrightarrow \S 9 \longrightarrow \S 10 \longrightarrow \S 11. \end{aligned}$$

Section 12 stands aside: it provides the optional source-energy supply consumed by Section 8 when their standalone forms are insufficient. All cross-references to Section 12 from earlier sections are forward references; that section is self-contained.

1.2. Master conditional theorem.

Theorem 1.1 (Master conditional). *Assume the same-point and cross-block jet-limit identities (Lemmas 3.1 and 3.2); the finite-scale whitening loss bound (Lemma 4.8); the toy anomaly visibility lower bound (Theorem 5.14); the two-point or four-point actual-zeta transverse suppression upper bound (Hypothesis 7.1 or Hypothesis 7.3); the corresponding local rigidity theorem (Theorem 8.3 or Theorem 8.6); the branch-exhaustion input (Hypothesis 8.1); the N -point odd-moment cancellation (Lemma 9.5); the aggregation step (Theorem 10.4); and the finite-height verification (Hypothesis 11.1). Then every nontrivial zero of ζ satisfies $\operatorname{Re} s = \frac{1}{2}$.*

Gap 1.2. The conditional statement is firm; its inputs are the section-level results listed above. The proof is short — a composition of the section-level results — and is written out in Section 11.

1.3. Notation and conventions. We use the symmetric pair convention $T \pm h$ (rather than $T \pm h/2$) throughout. The point-to-jet transform P_h of (2.7) is fixed by this choice. The scale parameter $Q = Q(\gamma)$ is a fixed slowly growing function of the height; all polynomial losses are stated as powers of Q with absolute exponents. For each height parameter T , I_T denotes an interval centered at T of length comparable to $Q(T)^{-1}$: there are fixed absolute constants $0 < c_I \leq C_I < \infty$ such that

$$c_I Q(T)^{-1} \leq |I_T| \leq C_I Q(T)^{-1}.$$

We also require $I_T \subset [T/2, 2T]$; high-height analysis takes T sufficiently large that this inclusion is automatic. The toy polynomial-weight subdomain in Section 5 uses the slow-scale condition $Q(T) = o(q(T))$, where $q = \theta'$; when that subdomain is not being invoked, the relevant lemmas state their window hypotheses explicitly. “Retained packets” refers to a polynomial-weight subfamily of the local packets at height γ on which the chart-regularity and Gram nondegeneracy hypotheses of Definition 2.4 hold.

2. LOCAL KERNEL AND JET NORMALIZATION

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This section establishes the local geometric machinery used throughout: the Riemann–Siegel phase chart $\Phi_T = \theta$ on the height window I_T of §1.3 (Definition 2.4), the phase kernel K_Φ (Definition 2.7), and the Gram-normalized point-to-jet transform P_h of (2.7).

2.1. The Riemann–Siegel phase chart.

Definition 2.1 (Riemann–Siegel phase). Let $L(z)$ be the unique holomorphic branch of $\log \Gamma z$ on $\Re z > 0$ for which $L(x) = \log \Gamma(x) \in \mathbb{R}$ when $x > 0$. For $t > 0$, define

$$(2.1) \quad \theta(t) := \Im L\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi,$$

and $Z(t) := e^{i\theta(t)} \zeta(1/2 + it)$. The functional equation gives

$$\zeta\left(\frac{1}{2} + it\right) = e^{-i\theta(t)} Z(t), \quad Z(t) \in \mathbb{R} \text{ for } t > 0.$$

Remark 2.2 (Branch convention for the Riemann–Siegel phase). The branch L in Definition 2.1 is the holomorphic logarithm of Γ on the half-plane $\Re z > 0$, not the principal logarithm applied pointwise to $\Gamma(z)$. This branch exists because Γ has no zeros or poles on $\Re z > 0$, and the half-plane is simply connected; the normalization on $(0, \infty)$ fixes the additive constant. Any other holomorphic logarithm on a connected subdomain would change θ only by an additive constant $2\pi k$, which leaves $q = \theta'$, the phase differences $\theta(x) - \theta(y)$, and therefore K_θ unchanged. A discontinuous branch, such as a principal-branch representative with jumps, is not part of the definition and cannot be used for derivative identities.

Lemma 2.3 (Differentiated theta asymptotics). *Uniformly for $t \in I_T$,*

$$(2.2) \quad q(t) := \theta'(t) = \frac{1}{2} \log \frac{t}{2\pi} - \frac{1}{48t^2} + O(t^{-4}),$$

$$(2.3) \quad q'(t) = \theta''(t) = \frac{1}{2t} + O(t^{-3}),$$

$$(2.4) \quad q''(t) = \theta'''(t) = -\frac{1}{2t^2} + O(t^{-4}).$$

The constants in the O -terms are absolute.

Proof. Let $z = 1/4 + it/2$. The half-plane $\Re z \geq 1/4$ lies in a fixed Stirling sector. By the differentiated form of Stirling’s expansion, after truncating $L(z)$ at z^{-3} the remainder and its t -derivatives up to order three are uniform in this sector, with the corresponding differentiated powers of $|z|$. Taking imaginary parts in (2.1) gives the standard Riemann–Siegel expansion

$$\theta(t) = \frac{t}{2} \log \frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + \frac{1}{48t} + O(t^{-3}),$$

with a remainder admitting three differentiations in t (needed for $q'' = \theta'''$). Termwise differentiation gives $q(t) = \theta'(t)$; differentiating once and twice more gives $q'(t)$ and $q''(t)$. Since $I_T \subset [T/2, 2T]$, these estimates are uniform on I_T , with absolute constants. $\square$

Definition 2.4 (Local phase chart). A retained high-height local phase chart is the interval I_T together with the real smooth phase

$$\Phi_T(t) := \theta(t) \quad (t \in I_T).$$

When T is fixed by context, $\Phi := \Phi_T$ and $q := \Phi'$. This is a specialized theta chart, not a general phase-chart axiom. The chart conditions used downstream are:

(P1) $\Phi_T = \theta$ on I_T ;

(P2) $q(t) \geq 1$ on I_T , after increasing the global lower-height cutoff if necessary.

If $Q(T) \geq 1$, then (P2) gives the polynomial-in- Q lower bound $q(t) \geq Q(T)^{-c_q}$ with $c_q = 0$. The uniform estimates $q(t) \asymp \log T$, $q'(t) = O(T^{-1})$, and $q''(t) = O(T^{-2})$ on I_T are supplied by Lemma 2.3.

Lemma 2.5 (Phase-derivative lower bound). *Condition (P2) of Definition 2.4 holds for all sufficiently large T . More quantitatively, uniformly on I_T ,*

$$q(t) \geq \frac{1}{2} \log \frac{T}{4\pi} - C T^{-2}$$

for an absolute constant C .

Proof. By Lemma 2.3, $q(t) = \frac{1}{2} \log(t/(2\pi)) + O(t^{-2})$. Since $t \in I_T \subset [T/2, 2T]$, $\log(t/(2\pi)) \geq \log(T/(4\pi))$, and the stated bound follows. The right-hand side is eventually larger than 1. $\square$

Remark 2.6 (No ζ -side regularity is used here). The chart uses ζ only through the identity $\zeta(1/2 + it) = e^{-i\theta(t)} Z(t)$ and through the theta phase. No lower bound on $|Z|$, no zero-free condition in I_T , and no source-energy input is used in Sections 2–3. Such conditions are imposed only in the later sections where actual-zeta transverse suppression or source-energy transfer is consumed.

2.2. The phase kernel.

Definition 2.7 (Phase kernel). For a real C^1 phase Φ on an interval I ,

$$(2.5) \quad K_\Phi(x, y) = \begin{cases} \frac{\sin(\Phi(x) - \Phi(y))}{\pi(x - y)}, & x \neq y, \\ \frac{q(x)}{\pi}, & x = y, \end{cases} \quad q = \Phi'.$$

Lemma 2.8 (Basic kernel properties). *K_Φ is real-valued and symmetric. If $\Phi \in C^1$, the extension across the diagonal is continuous and*

$$K_\Phi(T, T) = \frac{q(T)}{\pi}.$$

If $\Phi \in C^4$, the diagonal first and mixed second derivatives exist and are given by Lemma 2.10.

Proof. For $x \neq y$, symmetry follows from $\sin(\Phi(y) - \Phi(x))/(y - x) = \sin(\Phi(x) - \Phi(y))/(x - y)$. Real-valuedness is immediate. For joint continuity across the diagonal, near any $T \in I$ write

$$\Phi(x) - \Phi(y) = (x - y)A(x, y), \quad A(x, y) = \int_0^1 q(y + \tau(x - y)) d\tau.$$

The function A is continuous and $A(T, T) = q(T)$. For $x \neq y$,

$$K_\Phi(x, y) = \frac{A(x, y)}{\pi} \frac{\sin((x - y)A(x, y))}{(x - y)A(x, y)},$$

with the last factor interpreted by its continuous value 1 at zero. Hence $K_\Phi(x, y) \rightarrow q(T)/\pi = K_\Phi(T, T)$ whenever $(x, y) \rightarrow (T, T)$. $\square$

Lemma 2.9 (Off-diagonal kernel derivatives). *For $T_1 \neq T_2$ with $s = T_1 - T_2$, $\Delta = \Phi(T_1) - \Phi(T_2)$, $q_j = q(T_j)$, and $\Phi \in C^1$ near T_1, T_2 ,*

$$\begin{aligned} K_\Phi(T_1, T_2) &= \frac{\sin \Delta}{\pi s}, & K_x(T_1, T_2) &= \frac{q_1 s \cos \Delta - \sin \Delta}{\pi s^2}, \\ K_y(T_1, T_2) &= \frac{\sin \Delta - q_2 s \cos \Delta}{\pi s^2}, & K_{xy}(T_1, T_2) &= \frac{(q_1 + q_2) s \cos \Delta + (q_1 q_2 s^2 - 2) \sin \Delta}{\pi s^3}. \end{aligned}$$

Proof. Differentiate $K_\Phi(x, y) = \sin(\Phi(x) - \Phi(y))/\pi(x - y)$ by the quotient and chain rules, then evaluate at (T_1, T_2) . $\square$

Lemma 2.10 (Diagonal kernel derivatives). *If $\Phi \in C^4$ near T , then*

$$(2.6) \quad \begin{aligned} K_\Phi(T, T) &= \frac{q(T)}{\pi}, & K_x(T, T) &= K_y(T, T) = \frac{q'(T)}{2\pi}, \\ K_{xy}(T, T) &= \frac{q''(T) + 2q(T)^3}{6\pi}. \end{aligned}$$

Proof. Set $x = T + u$, $y = T + v$, and write $r = |u| + |v|$; derivatives below are evaluated at T . Factor the phase difference as

$$\Phi(T + u) - \Phi(T + v) = (u - v)A(u, v), \quad A(u, v) = \int_0^1 q(T + v + \tau(u - v)) d\tau,$$

where the integral form extends continuously to $u = v$. Taylor expansion of the integrand gives

$$A(u, v) = q + \frac{q'}{2}(u + v) + \frac{q''}{6}(u^2 + uv + v^2) + O(r^3).$$

For $u \neq v$, and then by continuous extension to $u = v$, $\Phi(T + u) - \Phi(T + v) = (u - v)A(u, v) = O(r)$ gives

$$\frac{\sin(\Phi(T + u) - \Phi(T + v))}{u - v} = A(u, v) - \frac{(u - v)^2}{6}A(u, v)^3 + O(r^3).$$

Keeping terms through total degree two gives

$$K_\Phi(T + u, T + v) = \frac{1}{\pi} \left[q + \frac{q'}{2}(u + v) + \left(\frac{q''}{6} - \frac{q^3}{6} \right) (u^2 + v^2) + \left(\frac{q''}{6} + \frac{q^3}{3} \right) uv + O(r^3) \right].$$

The coefficients of u , v , and uv are respectively $K_x(T, T)$, $K_y(T, T)$, and $K_{xy}(T, T)$, proving the claim. $\square$

2.3. Point-to-jet transform. For symmetric pairs $T \pm h$, define the Gram-normalized coalescing even/odd transform

$$(2.7) \quad P_h = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1/(2h) & 1/(2h) \end{pmatrix} = \text{diag}(\sqrt{2}, 1/\sqrt{2}) \begin{pmatrix} 1/2 & 1/2 \\ -1/(2h) & 1/(2h) \end{pmatrix}.$$

Thus P_h is the literal central value/derivative jet transform followed by the fixed diagonal rescaling $\text{diag}(\sqrt{2}, 1/\sqrt{2})$; at the level of Gram blocks this induces the corresponding diagonal congruence. The value/derivative basis produced by P_h is the *Gram-normalized basis*.

With this convention,

$$(2.8) \quad J(T) = \begin{pmatrix} 2K_\Phi(T, T) & \partial_y K_\Phi(T, T) \\ \partial_x K_\Phi(T, T) & \frac{1}{2} \partial_{xy} K_\Phi(T, T) \end{pmatrix},$$

which is the same-point block computed explicitly in Lemma 3.1.

Remark 2.11 (Pair-separation convention). The pair is $T - h, T + h$, with separation $2h$; the lower row of P_h carries $1/(2h)$. Any formula imported from the alternative $T \pm h/2$ convention or from the literal-jet matrix with lower row $(-1/h, 1/h)$ must be re-conjugated before use.

Remark 2.12 (Convention check). With the literal central value/derivative transform at the pair $T \pm h$, the same-point limit would be

$$\frac{1}{\pi} \begin{pmatrix} q(T) & \frac{1}{2}q'(T) \\ \frac{1}{2}q'(T) & \frac{1}{6}(q''(T) + 2q(T)^3) \end{pmatrix},$$

related to $J(T)$ by the diagonal congruence $\text{diag}(\sqrt{2}, 1/\sqrt{2})$ of (2.7); the $(1, 1)$ entry doubles and the $(2, 2)$ entry is halved. All same-point blocks, cross-blocks, finite-scale blocks, whitening matrices, and quotient covectors in this paper are read in the Gram-normalized convention.

3. JET-LIMIT LOCAL BLOCKS

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From the phase kernel K_Φ and the Gram-normalized transform P_h of Section 2, the small-separation limits of K_Φ at one and two centers produce the same-point Gram block $J(T)$ (Lemma 3.1) and the cross-block $N_{12}(T_1, T_2)$ (Lemma 3.2); for the theta phase, $J(T) \succ 0$ at sufficiently large height (Lemma 3.6).

3.1. Same-point jet limit.

Lemma 3.1 (Same-point jet limit). *Assume $\Phi \in C^5$ near T . Let*

$$A_h(T) = \begin{pmatrix} K_\Phi(T-h, T-h) & K_\Phi(T-h, T+h) \\ K_\Phi(T+h, T-h) & K_\Phi(T+h, T+h) \end{pmatrix}$$

with row and column order $(T-h, T+h)$. Then

$$P_h A_h(T) P_h^\top = J(T) + O(h^2) \quad (h \rightarrow 0),$$

where the $O(h^2)$ is entrywise, with a constant depending only on a fixed neighborhood of T and on a local C^5 -bound for Φ . The limiting block is

$$(3.1) \quad J(T) = \frac{1}{\pi} \begin{pmatrix} 2q(T) & \frac{1}{2}q'(T) \\ \frac{1}{2}q'(T) & \frac{1}{12}(q''(T) + 2q(T)^3) \end{pmatrix}.$$

Proof. Derivatives below are evaluated at T . Taylor expansion gives

$$\Phi(T \pm h) = \Phi(T) \pm hq + \frac{h^2}{2}q' \pm \frac{h^3}{6}q'' + \frac{h^4}{24}q''' + O(h^5),$$

so

$$\Phi(T-h) - \Phi(T+h) = -2hq - \frac{h^3}{3}q'' + O(h^5).$$

Using $\sin z = z - z^3/6 + O(z^5)$,

$$(3.2) \quad K_\Phi(T-h, T+h) = \frac{q}{\pi} + \frac{h^2}{6\pi}(q'' - 4q^3) + O(h^4).$$

Also $K_\Phi(T \pm h, T \pm h) = q(T \pm h)/\pi$, with

$$q(T \pm h) = q \pm hq' + \frac{h^2}{2}q'' \pm \frac{h^3}{6}q''' + O(h^4).$$

The $O(h^5)$ phase remainder and the $O(h^4)$ q -remainder are controlled by a local C^5 -bound for Φ ; after division by the separation $2h$, the off-diagonal kernel expansion has an $O(h^4)$ remainder, the order needed below since the $(2, 2)$ jet entry divides a second difference by h^2 . For symmetric M , conjugation by (2.7) gives

$$\begin{aligned} (P_h M P_h^\top)_{11} &= \frac{1}{2}(M_{11} + 2M_{12} + M_{22}), \\ (P_h M P_h^\top)_{12} &= \frac{1}{4h}(M_{22} - M_{11}), \\ (P_h M P_h^\top)_{22} &= \frac{1}{8h^2}(M_{11} - 2M_{12} + M_{22}). \end{aligned}$$

Substituting (3.2) and the Taylor series of $q(T \pm h)$: the constant and odd terms cancel in the second difference, while the first difference cancels its constant and even terms. The divisions by h and h^2 therefore leave entrywise $O(h^2)$ remainders:

$$\begin{aligned}(P_h A_h P_h^\top)_{11} &= \frac{2q}{\pi} + O(h^2), \\ (P_h A_h P_h^\top)_{12} &= \frac{q'}{2\pi} + O(h^2), \\ (P_h A_h P_h^\top)_{22} &= \frac{q'' + 2q^3}{12\pi} + O(h^2),\end{aligned}$$

which is (3.1). □

3.2. Cross-block jet limit.

Lemma 3.2 (Cross-block jet limit). *Assume $\Phi \in C^5$ in neighborhoods of T_1 and T_2 , let $T_1 \neq T_2$, set*

$$s = T_1 - T_2, \quad \Delta = \Phi(T_1) - \Phi(T_2), \quad q_j = q(T_j),$$

and assume $0 < h < |s|/3$, with h also sufficiently small that the four sampled points lie in the chosen neighborhoods of T_1 and T_2 . Define

$$C_h(T_1, T_2) = \begin{pmatrix} K_\Phi(T_1 - h, T_2 - h) & K_\Phi(T_1 - h, T_2 + h) \\ K_\Phi(T_1 + h, T_2 - h) & K_\Phi(T_1 + h, T_2 + h) \end{pmatrix}$$

with row order $(T_1 - h, T_1 + h)$ and column order $(T_2 - h, T_2 + h)$. Then

$$P_h C_h(T_1, T_2) P_h^\top = \frac{1}{\pi} N_{12}(T_1, T_2) + O(h^2) \quad (h \rightarrow 0),$$

where the $O(h^2)$ is entrywise for fixed $T_1 \neq T_2$; the implied constant may depend on a finite power of $|s|^{-1}$ and on local C^5 -bounds for Φ , but not on h . Here

$$(3.3) \quad N_{12} = \begin{pmatrix} \frac{2 \sin \Delta}{s} & \frac{\sin \Delta - q_2 s \cos \Delta}{s^2} \\ \frac{q_1 s \cos \Delta - \sin \Delta}{s^2} & \frac{(q_1 + q_2) s \cos \Delta + (q_1 q_2 s^2 - 2) \sin \Delta}{2s^3} \end{pmatrix}.$$

Proof. The kernel is smooth in a neighborhood of (T_1, T_2) , and the hypothesis $h < |s|/3$ gives $|s + u_1 - u_2| \geq |s|/3$ for $|u_1|, |u_2| \leq h$, bounding the Taylor remainder on the sampled square. Write $K_{ab} = \partial_x^a \partial_y^b K_\Phi(T_1, T_2)$. Taylor expansion to total order four gives, uniformly for $u_1, u_2 \in \{-h, +h\}$,

$$K_\Phi(T_1 + u_1, T_2 + u_2) = \sum_{a+b \leq 4} \frac{K_{ab}}{a! b!} u_1^a u_2^b + O_s(h^5).$$

The subscript indicates that the constant may depend on a finite power of $|s|^{-1}$ and on local C^5 -bounds for Φ in the chosen off-diagonal neighborhoods. For a general 2×2 matrix M ,

$$\begin{aligned}(P_h M P_h^\top)_{11} &= \frac{1}{2}(M_{11} + M_{12} + M_{21} + M_{22}), \\ (P_h M P_h^\top)_{12} &= \frac{1}{4h}(-M_{11} + M_{12} - M_{21} + M_{22}), \\ (P_h M P_h^\top)_{21} &= \frac{1}{4h}(-M_{11} - M_{12} + M_{21} + M_{22}), \\ (P_h M P_h^\top)_{22} &= \frac{1}{8h^2}(M_{11} - M_{12} - M_{21} + M_{22}).\end{aligned}$$

Equivalently, these four expressions apply the sign weights $1, \sigma_2, \sigma_1, \sigma_1 \sigma_2$ respectively to the four samples with $u_i = \sigma_i h$. Parity selects the required leading terms: the next surviving polynomial monomials in the $(1, 1)$ entry have total degree at least two, contributing $O(h^2)$; the $(1, 2)$ and $(2, 1)$ numerators select monomials with odd degree in one variable, so after division by h the surviving polynomial corrections are $O(h^2)$; the $(2, 2)$ numerator selects monomials odd in both variables, and division by h^2 again leaves $O(h^2)$. The Taylor remainder $O_s(h^5)$ per

sample contributes $O_s(h^5)$ to the $(1, 1)$ entry (no division), $O_s(h^4)$ to the off-diagonal entries (division by $4h$), and $O_s(h^3)$ to the $(2, 2)$ entry (division by $8h^2$). Hence

$$\begin{aligned} (P_h C_h P_h^\top)_{11} &= 2K_{00} + O(h^2), & (P_h C_h P_h^\top)_{12} &= K_{01} + O(h^2), \\ (P_h C_h P_h^\top)_{21} &= K_{10} + O(h^2), & (P_h C_h P_h^\top)_{22} &= \frac{1}{2}K_{11} + O(h^2). \end{aligned}$$

Substituting the off-diagonal derivative formulas of Lemma 2.9 gives (3.3). $\square$

Remark 3.3 (Fixed-separation scope). Lemma 3.2 is an off-diagonal jet limit at fixed separation. It is not a collision expansion for $T_1 \rightarrow T_2$; the constants are not asserted to be uniform as $s = T_1 - T_2 \rightarrow 0$. Any two-center blow-up or near-coalescence analysis must be derived from a separate scaling argument, rather than by citing this lemma with $|s|$ comparable to h .

3.3. Same-point Gram positivity.

Definition 3.4 (Same-point Gram determinant).

$$(3.4) \quad D_J(T) := 4q(T)^4 + 2q(T)q''(T) - 3q'(T)^2.$$

Lemma 3.5 (Algebraic same-point Gram criterion). *For $J(T)$ of (3.1),*

$$(3.5) \quad \text{Tr } J(T) = \frac{2q(T)^3 + 24q(T) + q''(T)}{12\pi}, \quad \det J(T) = \frac{D_J(T)}{12\pi^2}.$$

Moreover,

$$J(T) \succ 0 \iff q(T) > 0 \text{ and } D_J(T) > 0.$$

Proof. The trace and determinant follow by direct computation from (3.1). Since $J(T)$ is real symmetric, Sylvester's criterion gives positive definiteness exactly when its leading principal minor $2q(T)/\pi$ and its determinant are positive. $\square$

Lemma 3.6 (Same-point Gram positivity for the theta phase). *For $\Phi_T = \theta$, and for all sufficiently large retained heights, $J(t)$ is positive definite uniformly for $t \in I_T$. More precisely,*

$$(3.6) \quad \lambda_{\min}(J(t)) = \frac{2q(t)}{\pi} (1 + o_T(1)) = \frac{1}{\pi} \log \frac{t}{2\pi} (1 + o_T(1)) \quad (t \in I_T),$$

where the error is uniform on I_T . In particular this holds at the center $t = T$. After increasing the lower-height cutoff if necessary, $\lambda_{\min}(J(t)) \geq 1$ uniformly on I_T ; hence the polynomial-in- Q lower bound $\lambda_{\min}(J(t)) \geq Q(T)^{-C_J}$ holds with $C_J = 0$ whenever $Q(T) \geq 1$. The cutoff is qualitative; no explicit numerical lower height is claimed here.

Proof. Let $t \in I_T$. By Lemma 2.3, uniformly on I_T ,

$$q(t) = \frac{1}{2} \log(t/(2\pi)) + O(t^{-2}), \quad q'(t) = O(t^{-1}), \quad q''(t) = O(t^{-2}).$$

Therefore, uniformly on I_T ,

$$D_J(t) = 4q(t)^4 + O(q(t)t^{-2}) + O(t^{-2}) = 4q(t)^4 (1 + o_T(1)) > 0$$

for sufficiently large T , and $q(t) > 0$. By Lemma 3.5, $J(t) \succ 0$.

Let $0 < \lambda_1(t) \leq \lambda_2(t)$ be the eigenvalues of $J(t)$. Since $\lambda_2(t) \leq \text{Tr } J(t)$,

$$\lambda_1(t) = \frac{\det J(t)}{\lambda_2(t)} \geq \frac{\det J(t)}{\text{Tr } J(t)} = \frac{D_J(t)}{\pi(2q(t)^3 + 24q(t) + q''(t))} = \frac{2q(t)}{\pi} (1 + o_T(1)),$$

uniformly for $t \in I_T$. The Rayleigh quotient at e_1 gives $\lambda_1(t) \leq J_{11}(t) = 2q(t)/\pi$, proving (3.6). $\square$

Remark 3.7 (Scope of the positivity lemma). Lemma 3.6 is theta-specific. For an arbitrary real phase, the safe statement is only the algebraic criterion $q > 0$ and $D_J > 0$. This lemma does not prove any lower bound for unrelated anisotropy functionals such as $q'' - 6q + 2q^3$, nor does it supply source-energy input of the kind consumed in Section 12.

4. FINITE-SCALE LOCAL BLOCKS AND WHITENING

paper: [LIVE] v1: [migrated] referee: [in-progress] sympy: [verified] rh/sympy/sec-finite-scale-local-blocks-and-whitening
sim: [verified] rh/simulation/sec-finite-scale-local-blocks-and-whitening.py lean: [n/a]

Around a midpoint m at finite separation s , the same-point and cross-block jets of Section 3 give two same-point Gram blocks $G_{m,\pm}(s)$ and a mixed block $N_m(s)$. This section records the finite real-axis block package and the whitening-factor loss used by later comparison arguments.

Definition 4.1 (Finite packet domain). For a retained theta chart I_T , set $t_{\pm} = m \pm s/2$ and

$$\mathcal{D}_T := \{(m, s) \in \mathbb{R}^2 : t_- \in I_T, t_+ \in I_T\}.$$

Write $\mathcal{D}_T^{\times} := \{(m, s) \in \mathcal{D}_T : s \neq 0\}$. The orientation convention is that rows correspond to the center t_- and columns to the center t_+ . In applications $s > 0$, but allowing signed s makes the removable-singularity statement at $s = 0$ precise. Uniform estimates below are taken on $\mathcal{D}_T$, after the global lower-height cutoff has been enlarged if necessary.

4.1. Finite same-point and mixed blocks. For $(m, s) \in \mathcal{D}_T$, write

$$t_{\pm} = m \pm s/2, \quad q_{\pm} = q(t_{\pm}), \quad \Delta_m(s) = \Phi(t_-) - \Phi(t_+).$$

Definition 4.2 (Finite same-point and mixed blocks). For $(m, s) \in \mathcal{D}_T$, the finite same-point blocks are

$$(4.1) \quad G_{m,\pm}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_{\pm}) & \frac{1}{2}q'(t_{\pm}) \\ \frac{1}{2}q'(t_{\pm}) & \frac{1}{12}(q''(t_{\pm}) + 2q(t_{\pm})^3) \end{pmatrix}.$$

For $s \neq 0$, the finite mixed block is

$$(4.2) \quad N_m(s) = \frac{1}{\pi} \begin{pmatrix} -\frac{2 \sin \Delta_m(s)}{s} & \frac{\sin \Delta_m(s) + q_+ s \cos \Delta_m(s)}{s^2} \\ -\frac{\sin \Delta_m(s) + q_- s \cos \Delta_m(s)}{s^2} & \frac{(q_- + q_+) s \cos \Delta_m(s) + (2 - q_- q_+ s^2) \sin \Delta_m(s)}{2s^3} \end{pmatrix}.$$

Lemma 4.3 (Exact finite-block interpretation). For $(m, s) \in \mathcal{D}_T$,

$$G_{m,\pm}(s) = J(t_{\pm}).$$

For $(m, s) \in \mathcal{D}_T^{\times}$, moreover,

$$(4.3) \quad N_m(s) = \begin{pmatrix} 2K_{\Phi}(t_-, t_+) & \partial_y K_{\Phi}(t_-, t_+) \\ \partial_x K_{\Phi}(t_-, t_+) & \frac{1}{2} \partial_{xy} K_{\Phi}(t_-, t_+) \end{pmatrix}.$$

Equivalently, $N_m(s) = \pi^{-1} N_{12}$ for the matrix N_{12} of (3.3) taken at $(T_1, T_2) = (t_-, t_+)$, with $s_{12} = T_1 - T_2 = -s$ absorbed into the displayed signs in (4.2).

Proof. The identity $G_{m,\pm}(s) = J(t_{\pm})$ is immediate by comparing (4.1) with (3.1). For the mixed block, apply the off-diagonal kernel-derivative formulas of Lemma 2.9 at $(T_1, T_2) = (t_-, t_+)$; there $s_{12} = t_- - t_+ = -s$, $q_1 = q_-$, $q_2 = q_+$, and $\Delta = \Delta_m(s)$. Substitution gives the four entries of (4.3), which match (4.2) entry by entry. $\square$

Lemma 4.4 (Removable singularity of $N_m(s)$). Let $m \in I_T$. If $\Phi \in C^3$ on a neighborhood of m , then, as $s \rightarrow 0$ with $(m, s) \in \mathcal{D}_T^{\times}$, the entries of $N_m(s)$ of (4.2) extend continuously to $s = 0$ with

$$(4.4) \quad N_m(0) = J(m) = \frac{1}{\pi} \begin{pmatrix} 2q(m) & \frac{1}{2}q'(m) \\ \frac{1}{2}q'(m) & \frac{1}{12}(q''(m) + 2q(m)^3) \end{pmatrix}.$$

Proof. Write $q = q(m)$, $a = q'(m)$, $b = q''(m)$. Taylor expansion about m , using the symmetric points $m \pm s/2$, gives

$$\Delta_m(s) = \Phi(m - s/2) - \Phi(m + s/2) = -qs - \frac{b}{24}s^3 + o(s^3),$$

$$q_+ = q + \frac{a}{2}s + \frac{b}{8}s^2 + o(s^2), \quad q_- = q - \frac{a}{2}s + \frac{b}{8}s^2 + o(s^2).$$

Hence

$$\sin \Delta_m(s) = -qs + \left(\frac{q^3}{6} - \frac{b}{24}\right)s^3 + o(s^3), \quad \cos \Delta_m(s) = 1 - \frac{q^2}{2}s^2 + o(s^2).$$

Substituting into (4.2) entry by entry,

$$\begin{aligned} -\frac{2 \sin \Delta_m(s)}{s} &= 2q + o(1), \\ \frac{\sin \Delta_m(s) + q_+ s \cos \Delta_m(s)}{s^2} &= \frac{a}{2} + o(1), \\ -\frac{\sin \Delta_m(s) + q_- s \cos \Delta_m(s)}{s^2} &= \frac{a}{2} + o(1), \\ \frac{(q_- + q_+)s \cos \Delta_m(s) + (2 - q_- q_+ s^2) \sin \Delta_m(s)}{2s^3} &= \frac{b}{12} + \frac{q^3}{6} + o(1). \end{aligned}$$

The common factor $1/\pi$ in (4.2) gives (4.4). $\square$

Remark 4.5 (Coalescence is not the cross-block jet limit). Lemma 3.2 is a fixed-center-separation limit in the sampling parameter h ; its constants are not uniform as $s \rightarrow 0$ (Remark 3.3). Lemma 4.4 is a separate coalescence statement in the center-separation parameter s , proved by direct Taylor expansion above. From this point on, $N_m(0)$ denotes the continuous extension supplied by Lemma 4.4.

4.2. Whitened finite block.

Definition 4.6 (Whitened finite block). For $(m, s) \in \mathcal{D}_T$, with $N_m(0)$ understood by continuous extension, whenever $G_{m,-}(s)$ and $G_{m,+}(s)$ are positive definite, let $G_{m,\pm}(s)^{-1/2}$ denote the positive definite inverse square root and set

$$(4.5) \quad \hat{\Omega}_\zeta(s; m) := G_{m,-}(s)^{-1/2} N_m(s) G_{m,+}(s)^{-1/2}.$$

Corollary 4.7 (Coalescence: $\hat{\Omega}_\zeta(0; m) = I_2$). For every $m \in I_T$ at sufficiently large height, $\hat{\Omega}_\zeta(s; m)$ extends continuously to $s = 0$, along $(m, s) \in \mathcal{D}_T$, with $\hat{\Omega}_\zeta(0; m) = I_2$.

Proof. By Lemma 3.6, $J(m) \succ 0$ at every midpoint $m \in I_T$ at sufficiently large height. By continuity, $G_{m,\pm}(s) = J(m \pm s/2)$ remain positive definite for all sufficiently small $|s|$ with $(m, s) \in \mathcal{D}_T$, and the map $G \mapsto G^{-1/2}$ is continuous on the positive definite cone. Lemma 4.4 gives $N_m(0) = J(m)$, while $G_{m,\pm}(0) = J(m)$ by Lemma 4.3. Therefore $\hat{\Omega}_\zeta(0; m) = J(m)^{-1/2} J(m) J(m)^{-1/2} = I_2$. $\square$

4.3. Whitening loss.

Lemma 4.8 (Operator-norm bound for $G_{m,\pm}^{-1/2}$). For $\Phi_T = \theta$ on I_T , after enlarging the global lower-height cutoff if necessary, $G_{m,\pm}(s)$ are positive definite for every $(m, s) \in \mathcal{D}_T$, and there is a quantity $\varepsilon_T \rightarrow 0$, independent of $(m, s) \in \mathcal{D}_T$, such that

$$(4.6) \quad \|G_{m,\pm}(s)^{-1/2}\|_{\text{op}} \leq \sqrt{\frac{\pi}{2q(t_\pm)}} (1 + \varepsilon_T) \quad (T \rightarrow \infty).$$

After possibly enlarging the same lower-height cutoff,

$$(4.7) \quad \|G_{m,\pm}(s)^{-1/2}\|_{\text{op}} \leq Q(T)^{C_W}$$

uniformly on $\mathcal{D}_T$, with absolute exponent $C_W = 0$ whenever $Q(T) \geq 1$.

Proof. By Lemma 4.3, $G_{m,\pm}(s) = J(t_\pm)$; since $(m, s) \in \mathcal{D}_T$ implies $t_\pm \in I_T \subset [T/2, 2T]$, the theta-derivative estimates of Lemma 2.3 are uniform at $t = t_\pm$. The proof of Lemma 3.6 then applies with T replaced by an arbitrary $t \in I_T$, giving, uniformly on I_T ,

$$\lambda_{\min}(J(t)) = \frac{2q(t)}{\pi} (1 + o(1)).$$

Equivalently, for some $\delta_T \rightarrow 0$ independent of $t \in I_T$,

$$\lambda_{\min}(J(t)) \geq \frac{2q(t)}{\pi} (1 - \delta_T).$$

For large T , $1 - \delta_T > 0$; the finite same-point blocks are positive definite, and

$$\|G_{m,\pm}(s)^{-1/2}\|_{\text{op}} = \lambda_{\min}(G_{m,\pm}(s))^{-1/2} \leq \sqrt{\frac{\pi}{2q(t_{\pm})}} (1 - \delta_T)^{-1/2}.$$

Absorbing $(1 - \delta_T)^{-1/2}$ into $1 + \varepsilon_T$ gives (4.6).

For the polynomial bound, the uniform asymptotic above together with Lemma 2.5 gives $\lambda_{\min}(J(t)) \rightarrow \infty$ uniformly for $t \in I_T$; after enlarging the lower-height cutoff, $\lambda_{\min}(J(t)) \geq 1$ uniformly on I_T . Hence $\|G_{m,\pm}(s)^{-1/2}\|_{\text{op}} \leq 1 \leq Q(T)^0$ whenever $Q(T) \geq 1$, which is (4.7) with $C_W = 0$. $\square$

Remark 4.9 (Scope of Lemma 4.8). Lemma 4.8 controls only the real-axis same-point whitening factors attached to the theta phase. It does not prove actual-zeta transverse suppression, does not supply source-energy input, does not give a uniform bound on $\widehat{\Omega}_\zeta(s; m)$ itself beyond the whitening-factor loss, and does not give complex-disk or holomorphic whitening estimates. Any stronger whitening statement needed for a later complex-analytic argument must be introduced as a separate input or proved in a separate section.

4.4. Two-sided whitening ledger. The one-sided estimate of Lemma 4.8 prevents same-point whitening from enlarging upper bounds. Lower bounds require a separate ledger, because left and right multiplication by $G_{m,\pm}^{-1/2}$ can attenuate a nonzero matrix. Define the whitening map on 2×2 real matrices by

$$(4.8) \quad \mathbf{W}_{m,s}(X) := G_{m,-}(s)^{-1/2} X G_{m,+}(s)^{-1/2}.$$

Lemma 4.10 (Spectral size of the finite same-point blocks). *For $\Phi_T = \theta$, uniformly for $(m, s) \in \mathcal{D}_T$,*

$$(4.9) \quad \lambda_{\min}(G_{m,\pm}(s)) = \frac{2q(t_{\pm})}{\pi} (1 + o_T(1)),$$

$$(4.10) \quad \lambda_{\max}(G_{m,\pm}(s)) = \frac{q(t_{\pm})^3}{6\pi} (1 + o_T(1)).$$

Consequently, after increasing the global lower-height cutoff, there are absolute constants $c, C > 0$ such that, uniformly on $\mathcal{D}_T$,

$$(4.11) \quad cq(t_{\pm}) \leq \lambda_{\min}(G_{m,\pm}(s)) \leq Cq(t_{\pm}), \quad cq(t_{\pm})^3 \leq \lambda_{\max}(G_{m,\pm}(s)) \leq Cq(t_{\pm})^3.$$

Proof. By Lemma 4.3, $G_{m,\pm}(s) = J(t_{\pm})$. Since $(m, s) \in \mathcal{D}_T$, both $t_{\pm} \in I_T$, and the theta derivative asymptotics of Lemma 2.3 are uniform at these two points. The proof of Lemma 3.6 therefore applies uniformly with T replaced by any $t \in I_T$, giving (4.9).

For the largest eigenvalue, use

$$\lambda_{\max}(J(t)) = \text{Tr } J(t) - \lambda_{\min}(J(t)).$$

By (3.5), uniformly for $t \in I_T$,

$$\text{Tr } J(t) = \frac{2q(t)^3 + 24q(t) + q''(t)}{12\pi} = \frac{q(t)^3}{6\pi} (1 + o_T(1)),$$

while $\lambda_{\min}(J(t)) = O(q(t))$. Since $q(t) \rightarrow \infty$ uniformly on I_T , the trace term of size $q(t)^3$ dominates, which gives (4.10). The two-sided comparability bounds (4.11) follow by enlarging the lower-height cutoff once more. $\square$

Lemma 4.11 (Two-sided whitening ledger). *For $\Phi_T = \theta$, after increasing the global lower-height cutoff, there are absolute constants $c, C > 0$ such that for every $(m, s) \in \mathcal{D}_T$ and every real 2×2 matrix X ,*

$$(4.12) \quad c(q(t_-)q(t_+))^{-3/2} \|X\|_{\text{op}} \leq \|\mathbf{W}_{m,s}(X)\|_{\text{op}} \leq C(q(t_-)q(t_+))^{-1/2} \|X\|_{\text{op}}.$$

In particular, the lower-bound attenuation of two-sided whitening is at worst

$$(4.13) \quad (q(t_-)q(t_+))^{-3/2} \asymp (\log T)^{-3}$$

uniformly on $\mathcal{D}_T$.

Proof. Let

$$A = G_{m,-}(s)^{-1/2}, \quad B = G_{m,+}(s)^{-1/2}.$$

The upper bound follows from submultiplicativity:

$$\|AXB\|_{\text{op}} \leq \|A\|_{\text{op}} \|X\|_{\text{op}} \|B\|_{\text{op}} = \lambda_{\min}(G_{m,-})^{-1/2} \lambda_{\min}(G_{m,+})^{-1/2} \|X\|_{\text{op}}.$$

Using the lower-eigenvalue estimate in Lemma 4.10 gives the right-hand inequality in (4.12).

For the lower bound, write

$$X = A^{-1}(AXB)B^{-1}.$$

Therefore

$$\|X\|_{\text{op}} \leq \|A^{-1}\|_{\text{op}} \|AXB\|_{\text{op}} \|B^{-1}\|_{\text{op}} = \lambda_{\max}(G_{m,-})^{1/2} \lambda_{\max}(G_{m,+})^{1/2} \|AXB\|_{\text{op}}.$$

The upper-eigenvalue estimate in Lemma 4.10 gives

$$\|AXB\|_{\text{op}} \geq c (q(t_-)q(t_+))^{-3/2} \|X\|_{\text{op}},$$

which is the left-hand inequality in (4.12). Finally, Lemma 2.3 gives $q(t_{\pm}) \asymp \log T$ uniformly on I_T , and hence (4.13). $\square$

Corollary 4.12 (Polynomial ledger form). *Assume the packet scale has an upper phase-slope ledger*

$$(4.14) \quad q(t) \leq Q(T)^{C_q^+} \quad (t \in I_T)$$

for some absolute exponent C_q^+ . Then for all retained $(m, s) \in \mathcal{D}_T$ and all real 2×2 matrices X ,

$$(4.15) \quad \|W_{m,s}(X)\|_{\text{op}} \geq c Q(T)^{-3C_q^+} \|X\|_{\text{op}},$$

with an absolute constant $c > 0$. Thus any pre-whitening lower bound $\|X\|_{\text{op}} \geq Q(T)^{-A}$ becomes the post-whitening lower bound

$$(4.16) \quad \|W_{m,s}(X)\|_{\text{op}} \geq c Q(T)^{-(A+3C_q^+)}.$$

Proof. Combine the lower bound in Lemma 4.11 with (4.14) at $t_{\pm}$:

$$(q(t_-)q(t_+))^{-3/2} \geq Q(T)^{-3C_q^+}.$$

This gives (4.15); applying it to a matrix satisfying $\|X\|_{\text{op}} \geq Q(T)^{-A}$ gives (4.16). $\square$

Remark 4.13 (When the ledger is consumed). Lemma 4.11 is needed only when a toy, source-side, or finite-algebraic lower bound is first proved in unwhitened coordinates and then transported through $W_{m,s}$. If a visibility theorem is stated directly for the already whitened block $\widehat{\Omega}_{\zeta}$, then no additional whitening loss is charged. The same caveat applies to any fixed finite-dimensional norm other than $\|\cdot\|_{\text{op}}$: because all norms on $M_2(\mathbb{R})$ are equivalent, the constants in the ledger change, but the powers of $q(t_{\pm})$, $\log T$, and $Q(T)$ do not.

5. TOY ANOMALY AND TRANSVERSE OBSTRUCTION

paper: [LIVE] v1: [migrated] referee: [in-progress] sympy: [verified] rh/sympy/sec-toy-anomaly-and-transverse-obstruction.
sim: [verified] rh/simulation/sec-toy-anomaly-and-transverse-obstruction.py lean: [not-started]

The toy phase Φ_{toy} defined below is a real-analytic visibility model carrying an explicit even- u^2 deformation of the linear baseline, with $u = \beta - \frac{1}{2}$ the off-line displacement parameter. Applying the finite-block package of Section 4 to Φ_{toy} produces a toy whitened block $\widehat{\Omega}_{\text{toy}}(s; m, u, q_0)$. The off-diagonal trace $\mathcal{A}_{\text{toy}}$ of this block is identically zero on the on-line baseline and has nonzero leading- u^2 coefficient $F_{\text{toy}}(d, q_0)$, where $d = q_0 s$ is the dimensionless local separation. As $q_0 \rightarrow \infty$, this coefficient converges uniformly on compact d -intervals to a q_0 -independent

closed-form limit $F_\infty(d)$. The gauge-fixing projection Π_{trans} of Section 6 is a separate step; this section proves the scalar visibility input that such a projection must preserve.

5.1. Off-line toy phase.

Definition 5.1 (Off-line toy phase). Fix a height m_0 , a local frequency $q_0 > 0$, and an off-line displacement $u \in \mathbb{R}$. The off-line toy phase is the real-analytic phase

$$(5.1) \quad \Phi_{\text{toy}}(t; u, m_0, q_0) := q_0(t - m_0) + u^2(1 - \cos(q_0(t - m_0))), \quad t \in \mathbb{R}.$$

The displacement is identified with the off-line real part by $u = \beta - \frac{1}{2}$. By construction Φ_{toy} depends on u only through u^2 and is C^∞ on $\mathbb{R}$ for every (u, q_0) .

Remark 5.2 (Toy frequency convention). The parameter q_0 is part of the toy data. When it matters, we write

$$G_{\text{toy}, \pm}(s; m, u, q_0), \quad N_{\text{toy}}(s; m, u, q_0), \quad \widehat{\Omega}_{\text{toy}}(s; m, u, q_0).$$

In Section 5.4, where the toy is placed on a retained theta packet, the toy frequency is specialized to the local theta slope $q_0 = q(m)$. This convention is essential: the toy visibility theorem is uniform in an abstract high-frequency parameter q_0 , while the polynomial-weight subdomain chooses that parameter from the actual retained packet.

Lemma 5.3 (Toy phase derivatives). *With $\tau := t - m_0$, the derivatives of Φ_{toy} are*

$$(5.2) \quad q_{\text{toy}}(\tau) = q_0 + u^2 q_0 \sin(q_0 \tau), \quad q'_{\text{toy}}(\tau) = u^2 q_0^2 \cos(q_0 \tau), \quad q''_{\text{toy}}(\tau) = -u^2 q_0^3 \sin(q_0 \tau).$$

At $\tau = 0$, $q_{\text{toy}}(0) = q_0$, $q'_{\text{toy}}(0) = u^2 q_0^2$, $q''_{\text{toy}}(0) = 0$. The toy phase is not the retained theta chart of Definition 2.4; it is used only as a model phase. Its positivity analogue is

$$q_{\text{toy}}(\tau) \geq q_0(1 - u^2),$$

so the toy same-point blocks are safely in the positive-frequency regime whenever $|u| < 1$ and $q_0(1 - u^2) > 0$.

Proof. Direct differentiation of (5.1). The lower bound $q_{\text{toy}} \geq q_0(1 - u^2)$ follows from $|\sin| \leq 1$. $\square$

Lemma 5.4 (Symmetric-centre splitting at scale s). *With $t_\pm = m_0 \pm s/2$ and $d := q_0 s$, the toy phase derivatives at the symmetric centres satisfy*

$$(5.3) \quad q_\pm := q_{\text{toy}}(\pm s/2) = q_0 \pm u^2 q_0 \sin(d/2), \quad q'_\pm = u^2 q_0^2 \cos(d/2),$$

$$q''_\pm = \mp u^2 q_0^3 \sin(d/2).$$

In particular $q_+ - q_- = 2u^2 q_0 \sin(d/2)$, $q'_+ = q'_-$, and $q''_+ + q''_- = 0$. The phase difference $\Delta_m^{\text{toy}}(s)$ at the symmetric pair is

$$(5.4) \quad \Delta_m^{\text{toy}}(s) := \Phi_{\text{toy}}(t_-) - \Phi_{\text{toy}}(t_+) = -q_0 s = -d$$

(no u -dependence: the cosine is even at $\pm s/2$).

Proof. (5.3) substitutes $\tau = \pm s/2$ into (5.2) and uses oddness of $\sin$ and evenness of $\cos$. For (5.4), evaluate (5.1) at $\tau = \pm s/2$:

$$\Phi_{\text{toy}}(\pm s/2) = \pm \frac{q_0 s}{2} + u^2(1 - \cos(\mp d/2)) = \pm \frac{q_0 s}{2} + u^2(1 - \cos(d/2)),$$

and subtract. $\square$

5.2. Toy whitened block.

Definition 5.5 (Toy finite blocks). For $(m, s) \in \mathcal{D}_T$, with $t_{\pm} = m \pm s/2$, and for a frequency parameter $q_0 > 0$, evaluate the toy phase $\Phi_{\text{toy}}(t; u, m_0, q_0)$ of Definition 5.1 with centre $m_0 = m$. Set

$$(5.5) \quad G_{\text{toy}, \pm}(s; m, u, q_0) := G_{m, \pm}(s) \big|_{\Phi = \Phi_{\text{toy}}}, \quad N_{\text{toy}}(s; m, u, q_0) := N_m(s) \big|_{\Phi = \Phi_{\text{toy}}}.$$

Here the right-hand sides are the same-point and mixed blocks (4.1)–(4.2) of Section 4 evaluated with (q, q', q'', Δ_m) replaced by $(q_{\text{toy}}, q'_{\text{toy}}, q''_{\text{toy}}, \Delta_m^{\text{toy}})$ of Lemmas 5.3–5.4.

Definition 5.6 (Toy whitened block). For $(m, s) \in \mathcal{D}_T$, $q_0 > 0$, and $G_{\text{toy}, \pm}(s; m, u, q_0) \succ 0$, define

$$(5.6) \quad \widehat{\Omega}_{\text{toy}}(s; m, u, q_0) := G_{\text{toy}, -}(s; m, u, q_0)^{-1/2} N_{\text{toy}}(s; m, u, q_0) G_{\text{toy}, +}(s; m, u, q_0)^{-1/2}.$$

The inverse square root is the unique positive definite one.

Lemma 5.7 (Toy block positivity). *Fix a compact interval $D = [d_-, d_+] \subset (0, \infty)$. There exist $u_G = u_G(D) \in (0, \frac{1}{2}]$, $Q_G(D) \geq 1$, and $c_G(D) > 0$ such that whenever $d = q_0 s \in D$, $|u| \leq u_G$, and $q_0 \geq Q_G(D)$,*

$$G_{\text{toy}, \pm}(s; m, u, q_0) \succ 0, \quad \lambda_{\min}(G_{\text{toy}, \pm}(s; m, u, q_0)) \geq c_G(D) q_0.$$

In particular $\|G_{\text{toy}, \pm}(s; m, u, q_0)^{-1/2}\|_{\text{op}} \leq C_G(D) q_0^{-1/2}$ on the same region.

Proof. Write $r = u^2$, $z = \sin(d/2)$, and $c = \cos(d/2)$. From (5.3), after factoring out the powers of q_0 , the normalized same-point determinant has the form

$$q_0^{-4} D_J(t_{\pm}) = 4(1 \pm rz)^4 \mp 2rz(1 \pm rz) - 3r^2 c^2.$$

This expression converges uniformly to 4 as $r \rightarrow 0$, uniformly in $d \in D$. Decrease u_G , if necessary, so that $q_0^{-4} D_J(t_{\pm}) \geq 1$ and $q_{\pm} \geq q_0/2$ for $|u| \leq u_G$. By Lemma 3.5, $G_{\text{toy}, \pm} \succ 0$. Also, on $d \in D$ and $|u| \leq u_G$, $\text{Tr } G_{\text{toy}, \pm} \leq C(D) q_0^3$. Therefore

$$\lambda_{\min}(G_{\text{toy}, \pm}) \geq \frac{\det G_{\text{toy}, \pm}}{\text{Tr } G_{\text{toy}, \pm}} \geq c_G(D) q_0.$$

After increasing $Q_G(D)$, this holds throughout the retained high-frequency region. The inverse-square-root bound follows by taking reciprocals and square roots. $\square$

5.3. Toy anomaly functional and leading expansion.

Definition 5.8 (Toy anomaly functional). For a real 2×2 matrix $X = (x_{ij})$, the toy anomaly functional is the off-diagonal trace

$$(5.7) \quad \mathcal{A}_{\text{toy}}(X) := x_{12} + x_{21}.$$

Lemma 5.9 (Vanishing on the on-line baseline). *At $u = 0$ the toy phase reduces to the linear baseline $\Phi_{\text{toy}}(t; 0, m, q_0) = q_0(t - m)$, and $\widehat{\Omega}_{\text{toy}}(s; m, 0, q_0)$ is the constant-phase whitened block of the linear baseline. In particular*

$$(5.8) \quad \mathcal{A}_{\text{toy}}(\widehat{\Omega}_{\text{toy}}(s; m, 0, q_0)) = 0 \quad ((m, s) \in \mathcal{D}_T^{\times}).$$

Proof. At $u = 0$, (5.2) gives $q_{\pm} = q_0$ and $q'_{\pm} = q''_{\pm} = 0$; by (4.1), $G_{\text{toy}, \pm}(s; m, 0) = (1/\pi) \text{diag}(2q_0, q_0^3/6)$ is diagonal, and $G_{\text{toy}, -}^{-1/2} = G_{\text{toy}, +}^{-1/2}$ is the same diagonal matrix. By (4.2) with $q_+ = q_- = q_0$,

$$N_{\text{toy}}(s; m, 0, q_0)_{12} + N_{\text{toy}}(s; m, 0, q_0)_{21} = \frac{(q_+ - q_-) \cos \Delta_m^{\text{toy}}(s)}{\pi s} = 0.$$

Whitening by the diagonal $G^{-1/2}$ preserves this off-diagonal cancellation: $\widehat{\Omega}_{12} + \widehat{\Omega}_{21} = (G^{-1/2})_{11} (G^{-1/2})_{22} (N_{12} + N_{21}) = 0$. $\square$

Lemma 5.10 (Leading u^2 expansion of the off-diagonal trace). *Fix a compact interval $D = [d_-, d_+] \subset (0, \infty)$. There exist $Q_1 = Q_1(D) \geq 1$, $u_1 = u_1(D) > 0$, and a function $F_{\text{toy}} : D \times [Q_1, \infty) \rightarrow \mathbb{R}$ such that for every $0 < |u| < u_1$, every $q_0 \geq Q_1$, and every $(m, s) \in \mathcal{D}_T^\times$ with $d = q_0 s \in D$,*

$$(5.9) \quad \mathcal{A}_{\text{toy}}(\widehat{\Omega}_{\text{toy}}(s; m, u, q_0)) = u^2 F_{\text{toy}}(d, q_0) + u^4 R_{\text{toy}}(d, q_0, u),$$

with $|R_{\text{toy}}(d, q_0, u)| \leq C_R(D)$ uniformly on the region $\{|u| \leq u_1, d \in D, q_0 \geq Q_1(D)\}$. The high-frequency limit

$$(5.10) \quad F_\infty(d) := \lim_{q_0 \rightarrow \infty} F_{\text{toy}}(d, q_0)$$

exists, is reached uniformly on compact d -intervals, and admits the explicit closed form

$$(5.11) \quad F_\infty(d) = \frac{\sqrt{3}}{8d^3 \sin(d/2)} \left(2d^2 \cos d + 5d^2 \cos 2d - 7d^2 + 6d \sin d - 15d \sin 2d - 12 \cos 2d + 12 \right).$$

F_∞ is interpreted by removable continuation at the apparent $\sin(d/2)$ -poles $d = 2\pi k$, $k \geq 1$. Its zero set in $(0, \infty)$ is discrete. On every compact subinterval $D \subset (0, \infty)$ avoiding the zeros of this continued function there exist $Q_0(D) \geq 1$ and $c_{\text{toy}}(D) > 0$ with

$$(5.12) \quad |F_{\text{toy}}(d, q_0)| \geq c_{\text{toy}}(D) \quad (d \in D, q_0 \geq Q_0(D)).$$

Proof. Each entry of $G_{\text{toy}, \pm}(s; m, u, q_0)$ and $N_{\text{toy}}(s; m, u, q_0)$ is a real-analytic function of $r := u^2$ by (5.2) and (4.1)–(4.2). To make the uniformity in q_0 explicit after substituting $s = d/q_0$, use the natural powers q_0, q_0^2, q_0^3 attached to the value/derivative coordinates:

$$G_{11} = q_0 g_{11}, \quad G_{12} = q_0^2 g_{12}, \quad G_{22} = q_0^3 g_{22},$$

and similarly

$$N_{11} = q_0 n_{11}, \quad N_{12} = q_0^2 n_{12}, \quad N_{21} = q_0^2 n_{21}, \quad N_{22} = q_0^3 n_{22}.$$

The coefficients g_{ij} and n_{ij} are real-analytic functions of (r, d, η) , $\eta = q_0^{-1}$, on $[0, u_1^2] \times D \times [0, \eta_1]$, once u_1 is small and $Q_1 = \eta_1^{-1}$ is large. By Lemma 5.7, the normalized determinants of the $G_{\text{toy}, \pm}$ stay bounded away from zero on this compact parameter set. The explicit 2×2 positive-definite inverse-square-root formula therefore gives

$$(G_{\text{toy}, \pm}^{-1/2})_{11} = q_0^{-1/2} h_{11}^\pm, \quad (G_{\text{toy}, \pm}^{-1/2})_{12} = q_0^{-3/2} h_{12}^\pm, \quad (G_{\text{toy}, \pm}^{-1/2})_{22} = q_0^{-3/2} h_{22}^\pm,$$

where the $h_{ij}^\pm$ are again real-analytic and uniformly bounded on the same compact set. Multiplying the three factors in $G_{\text{toy}, -}^{-1/2} N_{\text{toy}} G_{\text{toy}, +}^{-1/2}$ shows that every entry of $\widehat{\Omega}_{\text{toy}}$, and in particular $\mathcal{A}_{\text{toy}}(\widehat{\Omega}_{\text{toy}})$, is real-analytic in r with uniformly bounded Taylor coefficients. Since the u^0 coefficient vanishes by Lemma 5.9, the leading nonzero term is at order u^2 ; call its coefficient $F_{\text{toy}}(d, q_0)$.

Direct computation using the closed-form 2×2 SPD inverse-square-root, applied to $G_{\text{toy}, \pm} = G_{\text{toy}, \pm}^{(0)} + u^2 G_{\text{toy}, \pm}^{(2)} + O(u^4)$ of (4.1) with the toy substitutions of (5.3) and to N_{toy} of Definition 5.5, yields an explicit expression for $F_{\text{toy}}(d, q_0)$ in $(\sin d, \cos d, \sin(d/2), \cos(d/2), q_0)$. After choosing $Q_1(D)$ large enough, this expression is regular for $d \in D$, $q_0 \geq Q_1(D)$, has a finite $q_0 \rightarrow \infty$ limit uniformly on D , and the limiting expression is (5.11).

The factor $\sin(d/2)$ of the denominator of (5.11) vanishes at $d = 2\pi k$, $k \in \mathbb{Z}_{\geq 1}$; the trigonometric numerator vanishes at the same points. Expanding the numerator at each $d = 2\pi k$ shows a factor of $\sin(d/2)$, so the apparent poles are removable and F_∞ extends real-analytically to $(0, \infty)$. On any compact subinterval avoiding the zeros of F_∞ , the difference $F_{\text{toy}}(d, q_0) - F_\infty(d)$ is $O(1/q_0)$ uniformly; for q_0 sufficiently large $|F_{\text{toy}}(d, q_0)| \geq \frac{1}{2}|F_\infty(d)|$, giving (5.12). The remainder R_{toy} is the u^4 -and-higher part of this real-analytic expansion. The compactness of the normalized parameter set and the uniform positive-definiteness just described give a Cauchy-type bound for these coefficients, uniformly for $d \in D$, $|u| \leq u_1$, and $q_0 \geq Q_1(D)$, after decreasing u_1 and increasing $Q_1(D)$ if necessary. $\square$

5.4. Toy retained subdomain and polynomial-weight density.

Definition 5.11 (Toy retained subdomain). Fix a compact interval $D = [d_-, d_+] \subset (0, \infty)$ avoiding the isolated zeros of F_∞ of Lemma 5.10. The signed toy retained subdomain at height T is

$$(5.13) \quad \mathcal{D}_T^{\text{toy}}(D) := \{(m, s) \in \mathcal{D}_T^\times : q(m) |s| \in D\}.$$

Thus the positive and negative separation branches are both retained. The comparison estimates below are stated for the absolute scalar functional $|\mathcal{A}_{\text{toy}}|$, so retaining both signs does not change the toy lower bound; see Lemma 5.13.

Lemma 5.12 (Polynomial-weight density of the toy retained subdomain). *Let*

$$q_T^- := \inf_{m \in I_T} q(m), \quad q_T^+ := \sup_{m \in I_T} q(m).$$

Assume $q_T^- > 0$ and

$$(5.14) \quad \frac{d_+}{q_T^-} \leq \frac{|I_T|}{4}.$$

Then

$$(5.15) \quad \frac{|\mathcal{D}_T^{\text{toy}}(D)|}{|\mathcal{D}_T|} \geq \frac{d_+ - d_-}{q_T^+ |I_T|}.$$

Consequently, if the retained windows satisfy

$$(5.16) \quad |I_T| \leq C_I Q(T)^{-1}, \quad q_T^+ \leq C_q q(T)$$

with absolute constants C_I, C_q , then

$$(5.17) \quad \frac{|\mathcal{D}_T^{\text{toy}}(D)|}{|\mathcal{D}_T|} \geq \frac{d_+ - d_-}{C_I C_q} \frac{Q(T)}{q(T)}.$$

The height hypothesis (5.14) is implied for all sufficiently large T in any scale regime with $|I_T| \geq c_I Q(T)^{-1}$, $q_T^- \geq c_q q(T)$, and $Q(T) = o(q(T))$, with constants depending only on D, c_I, c_q .

Proof. The finite packet domain is

$$\mathcal{D}_T = \{(m, s) : |m - T| + |s|/2 \leq |I_T|/2\},$$

a rotated square of area $|\mathcal{D}_T| = |I_T|^2$. Restrict to the central slab

$$S_T := \{m : |m - T| \leq |I_T|/4\}.$$

For $m \in S_T$, the $\mathcal{D}_T$ -constraint permits the full range

$$|s| \leq |I_T| - 2|m - T|.$$

In particular it contains the smaller symmetric range $|s| \leq |I_T|/4$. By (5.14),

$$\frac{d_+}{q(m)} \leq \frac{d_+}{q_T^-} \leq \frac{|I_T|}{4},$$

so both intervals

$$\left[\frac{d_-}{q(m)}, \frac{d_+}{q(m)} \right], \quad - \left[\frac{d_+}{q(m)}, \frac{d_-}{q(m)} \right]$$

are contained in the allowed s -range. Their combined length is $2(d_+ - d_-)/q(m)$, which is at least $2(d_+ - d_-)/q_T^+$. Therefore

$$|\mathcal{D}_T^{\text{toy}}(D)| \geq \int_{S_T} \frac{2(d_+ - d_-)}{q_T^+} dm = \frac{(d_+ - d_-)|I_T|}{q_T^+}.$$

Dividing by $|\mathcal{D}_T| = |I_T|^2$ proves (5.15). The estimate (5.17) follows immediately from (5.16). Finally, if $|I_T| \geq c_I Q(T)^{-1}$ and $q_T^- \geq c_q q(T)$, then (5.14) follows from

$$\frac{d_+}{c_q q(T)} \leq \frac{c_I}{4Q(T)}, \quad \text{i.e.} \quad \frac{Q(T)}{q(T)} \leq \frac{c_I c_q}{4d_+},$$

which holds eventually whenever $Q(T) = o(q(T))$. $\square$

Lemma 5.13 (Signed-separation invariance of the scalar toy channel). *For the toy finite blocks of Definition 5.5, with fixed $q_0 > 0$,*

$$\widehat{\Omega}_{\text{toy}}(-s; m, u, q_0) = \widehat{\Omega}_{\text{toy}}(s; m, u, q_0)^\top$$

whenever both sides are defined. Consequently

$$(5.18) \quad \mathcal{A}_{\text{toy}}(\widehat{\Omega}_{\text{toy}}(-s; m, u, q_0)) = \mathcal{A}_{\text{toy}}(\widehat{\Omega}_{\text{toy}}(s; m, u, q_0)).$$

Proof. Changing s to $-s$ swaps the two centers t_- and t_+ . The same-point blocks therefore satisfy

$$G_{\text{toy},-}(-s; m, u, q_0) = G_{\text{toy},+}(s; m, u, q_0), \quad G_{\text{toy},+}(-s; m, u, q_0) = G_{\text{toy},-}(s; m, u, q_0),$$

and the mixed block satisfies $N_{\text{toy}}(-s; m, u, q_0) = N_{\text{toy}}(s; m, u, q_0)^\top$, by the symmetry of K_Φ and the row/column convention of Definition 4.2. Positive definite inverse square roots commute with transposition for symmetric matrices, so the displayed transpose identity for $\widehat{\Omega}_{\text{toy}}$ follows. Since $\mathcal{A}_{\text{toy}}(X) = X_{12} + X_{21}$ is invariant under transpose, (5.18) follows. $\square$

5.5. Toy anomaly visibility.

Theorem 5.14 (Toy anomaly visibility). *Let $D \subset (0, \infty)$ be a compact interval avoiding the isolated zeros of F_∞ of Lemma 5.10. There are constants $Q_{\text{toy}}(D) \geq 1$ and $u_{\text{toy}}(D) > 0$ such that, for every off-line displacement $u = \beta - \frac{1}{2} \in \mathbb{R}$ with $0 < |u| \leq u_{\text{toy}}(D)$, every $q_0 \geq Q_{\text{toy}}(D)$, and every $(m, s) \in \mathcal{D}_T^\times$ with $d = q_0 s \in D$, the toy whitened block $\widehat{\Omega}_{\text{toy}}(s; m, u, q_0)$ of Definition 5.6 satisfies*

$$(5.19) \quad |\mathcal{A}_{\text{toy}}(\widehat{\Omega}_{\text{toy}}(s; m, u, q_0))| \geq \frac{1}{2} c_{\text{toy}}(D) u^2.$$

The constant $c_{\text{toy}}(D) > 0$ is the lower bound of Lemma 5.10 and depends on D only. On the polynomial-weight subset of midpoints and separations on which $d \in D$, the visibility exponent is $A_{\text{toy}} = 0$: the lower bound is $(c_{\text{toy}}(D)/2) u^2 Q^0$, independent of the height.

Proof. Choose $Q_{\text{toy}}(D) \geq \max\{Q_1(D), Q_G(D)\}$. Let $u_G(D)$ be the smallness threshold from Lemma 5.7. Choose

$$u_{\text{toy}}(D) \leq \min\{u_1(D), u_G(D)\}$$

small enough that both lemmas apply. By (5.9) and (5.12),

$$|\mathcal{A}_{\text{toy}}(\widehat{\Omega}_{\text{toy}}(s; m, u, q_0))| \geq u^2 (c_{\text{toy}}(D) - u^2 C_R(D)).$$

Decreasing $u_{\text{toy}}(D)$ so that $u_{\text{toy}}(D)^2 C_R(D) \leq c_{\text{toy}}(D)/2$ gives (5.19). $\square$

Corollary 5.15 (Visibility on the polynomial-weight subdomain). *Fix D and the constants $Q_{\text{toy}}(D)$, $u_{\text{toy}}(D)$, $c_{\text{toy}}(D)$ of Theorem 5.14. Assume the height is large enough that $q_T^- \geq Q_{\text{toy}}(D)$, where $q_T^- := \inf_{m \in I_T} q(m)$. On the toy retained subdomain we set $q_0 = q(m)$. Then for every $0 < |u| \leq u_{\text{toy}}(D)$ and every $(m, s) \in \mathcal{D}_T^{\text{toy}}(D)$,*

$$(5.20) \quad |\mathcal{A}_{\text{toy}}(\widehat{\Omega}_{\text{toy}}(s; m, u, q(m)))| \geq \frac{1}{2} c_{\text{toy}}(D) u^2.$$

If, in addition, the hypotheses of Lemma 5.12 hold, then this lower bound holds on a subset of $\mathcal{D}_T$ of relative density at least the right-hand side of (5.15), and hence at least (5.17) under (5.16).

Proof. Let $(m, s) \in \mathcal{D}_T^{\text{toy}}(D)$. Then $d = q(m)|s| \in D$ and $q_0 = q(m) \geq q_T^- \geq Q_{\text{toy}}(D)$. If $s > 0$, Theorem 5.14 gives (5.20). If $s < 0$, apply the theorem to $|s|$ and then use Lemma 5.13. The density statement is Lemma 5.12. $\square$

Remark 5.16 (Transverse projection deferred). The functional $\mathcal{A}_{\text{toy}}$ annihilates the on-line baseline (Lemma 5.9). The later projection Π_{trans} must be chosen so that this scalar functional descends to or is represented on the retained transverse quotient. Once Section 6 proves that compatibility, the visibility bound (5.19) transfers to the Π_{trans} -projected comparison with the same exponent $A_{\text{toy}} = 0$. This compatibility is not proved in the present section.

Remark 5.17 (Compatibility with the archived v1 toy matrix). The toy phase (5.1) is a replacement toy for the clean scalar-functional rebuild. It is not the archived v1 matrix $M(d)$, and this section does not import the v1 calibration map $\Psi_{x,d}$ or the Frobenius pairings with A_{val} . Consequently this change is isolated for the present rebuild through Sections 6–8, where only the scalar functional $\mathcal{A}_{\text{toy}}$, the exponent $A_{\text{toy}} = 0$, and the lower bound (5.19) are consumed. Any later attempt to import a v1 theorem that explicitly uses $M(d)$, $\Phi_K(M(d))$, A_{val} , or $\Psi_{x,d}$ must be rederived for the present toy phase or kept in the archive.

Remark 5.18 (No uniform extension; polynomial-weight scope). The visibility lower bound (5.19) cannot be extended to all of $\mathcal{D}_T$. At $s = 0$, the same coalescence argument as Corollary 4.7, applied to the toy finite blocks, gives $\hat{\Omega}_{\text{toy}}(0; m, u, q_0) = I_2$ for every u . Hence $\mathcal{A}_{\text{toy}}(\hat{\Omega}_{\text{toy}}(0; m, u, q_0)) = 0$; no u^2 -lower bound can hold at the coalescence locus $d = 0$. The regimes $d \rightarrow \infty$ and d approaching an isolated zero of F_∞ are excluded by the compactness and avoidance hypotheses on D in Definition 5.11. The signed polynomial-weight subdomain $\mathcal{D}_T^{\text{toy}}(D)$ carries the toy visibility lower bound by Corollary 5.15; its relative density is polynomial in Q by Lemma 5.12 and is the only density loss contributed by this replacement of the impossible all-of- $\mathcal{D}_T$ assertion.

6. TRANSVERSE PROJECTION

paper: [LIVE] v1: [migrated] referee: [in-progress] sympy: [verified] rh/sympy/sec-transverse-projection.py
sim: [verified] rh/simulation/sec-transverse-projection.py lean: [not-started]

The scalar toy anomaly functional $\mathcal{A}_{\text{toy}}$ of Definition 5.8 fixes the two-point value subspace as its kernel. The transverse projection in this section is the Hilbert–Schmidt orthogonal projection of the Gram-normalized 2×2 matrix space onto the one-dimensional complement of that kernel. This gives a canonical two-point Π_{trans} compatible with the scalar visibility lower bound of Theorem 5.14. The projection has Hilbert–Schmidt operator norm one, contributes no polynomial loss, and allows the actual-zeta two-point suppression statement of Section 7 to be formulated in the same scalar channel as the toy lower bound. All occurrences of Π_{trans} in this section are therefore understood as two-point objects on $M_2(\mathbb{R})$. This section does not define the four-point projection $\Pi_{\text{trans}}^{(4)}$.

6.1. Value subspace and transverse projection. We work on $M_2(\mathbb{R})$ with the Hilbert–Schmidt inner product $\langle X, Y \rangle_{\text{HS}} := \text{tr}(X^\top Y)$.

Definition 6.1 (Value subspace). The two-point scalar value subspace is

$$(6.1) \quad \mathcal{V}_{\text{val}} := \{X \in M_2(\mathbb{R}) : x_{12} + x_{21} = 0\} = \ker \mathcal{A}_{\text{toy}}.$$

Lemma 6.2 (Explicit basis for $\mathcal{V}_{\text{val}}$). $\mathcal{V}_{\text{val}}$ is a three-dimensional subspace of $M_2(\mathbb{R})$ with orthonormal basis

$$(6.2) \quad e_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad e_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad e_{\text{anti}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Its one-dimensional orthogonal complement is

$$(6.3) \quad \mathcal{V}_{\text{val}}^\perp = \mathbb{R} e_{\text{sym}}, \quad e_{\text{sym}} := \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Proof. The four matrices $e_{11}, e_{22}, e_{\text{anti}}, e_{\text{sym}}$ form an orthonormal basis of $M_2(\mathbb{R})$ by direct calculation. The first three have $x_{12} + x_{21} = 0$, so they lie in $\mathcal{V}_{\text{val}}$. The matrix e_{sym} has $x_{12} + x_{21} = \sqrt{2} \neq 0$ and is orthogonal to the first three, so it spans $\mathcal{V}_{\text{val}}^\perp$. $\square$

Remark 6.3 (Four-point projection not supplied here). This section defines the two-center projection on $M_2(\mathbb{R})$. The four-center projection $\Pi_{\text{trans}}^{(4)}$ appearing in Hypothesis 7.3 and Theorem 8.6 is a separate object. No four-point quotient, four-point value subspace, or four-point scalar functional is proved in this section.

Definition 6.4 (Two-point transverse projection). The two-point transverse projection $\Pi_{\text{trans}} : M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$ is the Hilbert–Schmidt orthogonal projection onto $\mathcal{V}_{\text{val}}^\perp$. Equivalently,

$$(6.4) \quad \Pi_{\text{trans}} X = \frac{x_{12} + x_{21}}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Definition 6.5 (Two-point scalar transverse channel). For a real two-center whitened block $X \in M_2(\mathbb{R})$, define

$$(6.5) \quad \mathcal{A}_2(X) := \mathcal{A}_{\text{toy}}(\Pi_{\text{trans}} X).$$

Lemma 6.8 gives $\mathcal{A}_2(X) = \mathcal{A}_{\text{toy}}(X) = x_{12} + x_{21}$. Thus $\mathcal{A}_2$ is a scalar functional on two-point matrix representatives, or equivalently on the quotient $M_2(\mathbb{R})/\mathcal{V}_{\text{val}}$ after choosing the Hilbert–Schmidt orthogonal representative $\Pi_{\text{trans}} X$.

6.2. Properties of Π_{trans} .

Lemma 6.6 (Projection properties). *The operator Π_{trans} is idempotent, self-adjoint with respect to the Hilbert–Schmidt inner product, has kernel exactly $\mathcal{V}_{\text{val}}$, and, as an operator on $(M_2(\mathbb{R}), \|\cdot\|_{\text{HS}})$,*

$$(6.6) \quad \|\Pi_{\text{trans}}\|_{\text{op,HS} \rightarrow \text{HS}} = 1.$$

Proof. Formula (6.4) is the same as $\Pi_{\text{trans}} X = \langle X, e_{\text{sym}} \rangle_{\text{HS}} e_{\text{sym}}$. Hence Π_{trans} is the Hilbert–Schmidt orthogonal projection onto the nonzero subspace $\mathbb{R}e_{\text{sym}}$. Idempotence, self-adjointness, kernel $\mathcal{V}_{\text{val}}$, and operator norm one follow immediately. $\square$

Lemma 6.7 (Quotient representative and scalar norm). *For every $X \in M_2(\mathbb{R})$,*

$$(6.7) \quad \Pi_{\text{trans}} X = \frac{\mathcal{A}_{\text{toy}}(X)}{\sqrt{2}} e_{\text{sym}}, \quad X - \Pi_{\text{trans}} X \in \mathcal{V}_{\text{val}},$$

and

$$(6.8) \quad \|\Pi_{\text{trans}} X\|_{\text{HS}}^2 = \frac{|\mathcal{A}_{\text{toy}}(X)|^2}{2}.$$

Proof. The representative identity follows from (6.4) and $e_{\text{sym}} = 2^{-1/2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Applying $\mathcal{A}_{\text{toy}}$ to $X - \Pi_{\text{trans}} X$ gives zero by Lemma 6.8, so the residual lies in $\mathcal{V}_{\text{val}}$. The norm identity follows from $\|e_{\text{sym}}\|_{\text{HS}} = 1$. $\square$

Lemma 6.8 (Scalar-channel compatibility). *For every $X \in M_2(\mathbb{R})$,*

$$(6.9) \quad \mathcal{A}_{\text{toy}}(\Pi_{\text{trans}} X) = \mathcal{A}_{\text{toy}}(X).$$

Consequently the toy visibility lower bound (5.19) is unchanged if the toy whitened block is first replaced by its projected representative $\Pi_{\text{trans}} \hat{\Omega}_{\text{toy}}$. No matrix-norm lower bound is asserted; only the scalar channel $\mathcal{A}_2 = \mathcal{A}_{\text{toy}} \circ \Pi_{\text{trans}}$ is transported.

Proof. By (6.4), $(\Pi_{\text{trans}} X)_{12} = (\Pi_{\text{trans}} X)_{21} = (x_{12} + x_{21})/2$. Therefore

$$\mathcal{A}_{\text{toy}}(\Pi_{\text{trans}} X) = \frac{x_{12} + x_{21}}{2} + \frac{x_{12} + x_{21}}{2} = x_{12} + x_{21} = \mathcal{A}_{\text{toy}}(X).$$

The final assertion follows by applying this equality to $X = \hat{\Omega}_{\text{toy}}(s; m, u, q_0)$ inside the absolute value in (5.19). $\square$

Lemma 6.9 (Two-point scalar quotient well-definedness). *The scalar channel $\mathcal{A}_2$ descends to the quotient $M_2(\mathbb{R})/\mathcal{V}_{\text{val}}$. Equivalently, if $X - Y \in \mathcal{V}_{\text{val}}$, then*

$$(6.10) \quad \mathcal{A}_2(X) = \mathcal{A}_2(Y).$$

Moreover, for every $X \in M_2(\mathbb{R})$,

$$(6.11) \quad \mathcal{A}_2(X) = \mathcal{A}_{\text{toy}}(\Pi_{\text{trans}} X) = \mathcal{A}_{\text{toy}}(X) = \sqrt{2} \langle X, e_{\text{sym}} \rangle_{\text{HS}}.$$

Proof. If $X - Y \in \mathcal{V}_{\text{val}} = \ker \mathcal{A}_{\text{toy}}$, then $\mathcal{A}_{\text{toy}}(X) - \mathcal{A}_{\text{toy}}(Y) = 0$. By Definition 6.5 and Lemma 6.8, $\mathcal{A}_2 = \mathcal{A}_{\text{toy}}$ on matrix representatives; hence (6.10). The identities in (6.11) follow from Lemma 6.8 and $\langle X, e_{\text{sym}} \rangle_{\text{HS}} = (x_{12} + x_{21})/\sqrt{2}$. $\square$

6.3. Scope of the chosen projection.

Remark 6.10 (No richer quotient is used in the scalar rebuild). The downstream two-point comparison consumes only the scalar visibility $|\mathcal{A}_2(\widehat{\Omega}_{\text{toy}})|$ and the matching actual-zeta scalar upper bound $|\mathcal{A}_2(\widehat{\Omega}_\zeta)|$. At this scalar level, the kernel of $\mathcal{A}_{\text{toy}}$ is the complete value subspace. The orthogonal projection of Definition 6.4 gives the canonical Hilbert–Schmidt representative of the quotient map $M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})/\mathcal{V}_{\text{val}}$, with no polynomial loss. Richer matrix-level quotient constructions from the archived local-projective-rigidity machinery are not imported here; cf. Remark 5.17.

Remark 6.11 (Residual gauge). The retained chart of Definition 2.4 pins $\Phi_T = \theta$. Additive constants $\Phi \mapsto \Phi + c$ leave q, q', q'' , hence $G_{m,\pm}$, N_m , and $\widehat{\Omega}_\zeta$, unchanged. Affine-jet shifts $\Phi \mapsto \Phi + at$ change q to $q + a$ and do not preserve the chart class. Thus there is no residual gauge action on the two-point matrix representatives beyond the trivial additive-constant invariance.

Lemma 6.12 (Sufficient two-point projection input). *The transverse projection Π_{trans} and scalar channel $\mathcal{A}_2$ supply the two-point projection input required by Section 8: the toy visibility lower bound (5.19) transports through Π_{trans} by Lemma 6.8, and the Hilbert–Schmidt operator-norm bound (6.6) contributes no polynomial-in- Q loss. The actual-zeta two-point upper bound must be stated in the same scalar channel, namely as a bound for $|\mathcal{A}_2(\widehat{\Omega}_\zeta(s; m))|$.*

Proof. Combine Lemmas 6.6, 6.7, 6.8, and 6.9 with Theorem 5.14 and Corollary 5.15. $\square$

7. ACTUAL-ZETA TRANSVERSE SUPPRESSION

paper: [LIVE] v1: [not-started] referee: [not-started] sympy: [not-started] sim: [not-started] lean: [not-started]

The actual zeta phase Φ defines two whitened blocks — the two-center block $\widehat{\Omega}_\zeta(s; m)$ and a four-center analogue $\widehat{\Omega}_\zeta^{(4)}$ for the symmetric quartet — on which the actual-zeta transverse anomaly is suppressed at a strictly larger polynomial exponent than the corresponding toy lower bound of Theorem 5.14. In the two-point case, the transverse scalar channel is the one fixed in Section 6:

$$\mathcal{A}_2(X) := \mathcal{A}_{\text{toy}}(\Pi_{\text{trans}}X) = \mathcal{A}_{\text{toy}}(X).$$

The equality is Lemma 6.8. Thus Π_{trans} acts on the block before the scalar functional is applied; the expression $\Pi_{\text{trans}}\mathcal{A}_2(X)$ is not used.

The two-point and four-point comparisons of Section 8 consume different forms of this suppression, stated separately below.

7.1. Two-point suppression.

Hypothesis 7.1 (Two-point actual-zeta transverse suppression). For the two-center packet associated with ρ and a symmetry partner, after the gauge-fixed normalization of Definition 2.4,

$$(7.1) \quad |\mathcal{A}_2(\widehat{\Omega}_\zeta(s; m))| \leq Q^{-B_{\zeta,2}}$$

for an absolute exponent $B_{\zeta,2} > A_{\text{toy}}$ on the retained subfamily matched to the toy visibility theorem. Equivalently, by Lemma 6.8,

$$|\mathcal{A}_{\text{toy}}(\Pi_{\text{trans}}\widehat{\Omega}_\zeta(s; m))| = |\mathcal{A}_{\text{toy}}(\widehat{\Omega}_\zeta(s; m))| \leq Q^{-B_{\zeta,2}}.$$

Gap 7.2. The two-point case of the structural gap. Consumed by Theorem 8.3. Admissible proof routes: (i) a standalone route from the local jet-Gram geometry alone; (ii) an energy-supply route consuming the source bound of Section 12 via Lemma 12.5. v1 prose to mine: the odd holomorphic transverse channel and corrected finite- s holomorphic whitening sections.

7.2. Four-point suppression.

Hypothesis 7.3 (Four-point actual-zeta transverse suppression). For the four-center packet associated with the symmetry quartet $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$, let $\Pi_{\text{trans}}^{(4)}$ be a four-point transverse projection and let $\mathcal{A}_{4,\text{trans}}$ be the corresponding four-point transverse functional, i.e. a scalar or finite-dimensional normed output after the four-point projection has been applied. Then

$$(7.2) \quad \|\mathcal{A}_{4,\text{trans}}(\widehat{\Omega}_\zeta^{(4)})\| \leq Q^{-B_{\zeta,4}}$$

for an absolute exponent $B_{\zeta,4} > A_{\text{toy}}^{(4)}$ on the retained four-point subfamily.

Gap 7.4. The four-point case of the structural gap. Consumed by Theorem 8.6. Same admissible proof routes as Hypothesis 7.1. The projection $\Pi_{\text{trans}}^{(4)}$, the four-point value subspace, and the functional $\mathcal{A}_{4,\text{trans}}$ are not supplied by Section 6; they are separate four-center inputs. v1 prose to mine: the local projective rigidity of the one-pair and multi-pair families section, four-point variant.

Remark 7.5 (Standalone and energy-supply routes). Hypotheses 7.1 and 7.3 are stated independently of the technology of their proofs. A purely local proof would use only the jet-Gram and quotient geometry of Sections 3–6; an energy-supply route may use the source-energy module of Section 12. Forward references from Section 8 indicate which hypothesis is invoked, not which route proves it.

8. LOCAL PROJECTIVE RIGIDITY

paper: [LIVE] v1: [not-started] referee: [not-started] sympy: [not-started] sim: [not-started] lean: [not-started]

Theorem 5.14 provides the toy lower bound and Hypotheses 7.1 and 7.3 provide the matching actual-zeta upper bounds on the transverse anomaly under Π_{trans} of Definition 6.4. The present section assembles the comparison. On the two-point family attached to ρ and a single symmetry partner, and on the four-point family of the full symmetric quartet $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$, the exponent inequality $A_{\text{toy}} < B_\zeta$ (resp. its four-point analogue $A_{\text{toy}}^{(4)} < B_\zeta^{(4)}$) is incompatible with the existence of the configuration. The two cases share the comparison mechanism but differ in the underlying jet-Gram object: the two-point case uses the cross-block N_{12} of Lemma 3.2; the four-point case uses a four-center analogue $\widehat{\Omega}_\zeta^{(4)}$ and a four-point extension $\Pi_{\text{trans}}^{(4)}$ of the projection of Definition 6.4. Either case suffices for the aggregation step in Section 10.

8.1. Branch exhaustion.

Hypothesis 8.1 (Branch exhaustion). Every admissible off-line packet produced by the branch-entry construction is excluded by at least one of:

- (i) the two-point scalar branch of Theorem 8.3; or
- (ii) the four-point full-quartet branch of Theorem 8.6.

Remark 8.2 (Status of Hypothesis 8.1). Hypothesis 8.1 is not a consequence of Definition 6.4 and is not implied by Sections 6–7. The master conditional theorem of Section 1.2 consumes Hypothesis 8.1 alongside Theorems 8.3 and 8.6.

8.2. Two-point comparison. The two-point comparison takes ρ and one symmetry partner (typically $\bar{\rho}$ or $1 - \rho$) and reads off the joint contribution to $N_m(s)$ at matched midpoint and separation. The exponent inequality $A_{\text{toy}} < B_\zeta$ excludes the pair.

Theorem 8.3 (Two-point rigidity). *Let $\rho = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$ and $|\gamma|$ sufficiently large, and let (T_1, T_2) be the height pair attached to ρ and one of its symmetry partners. Under Lemmas 3.1, 3.2, 4.8, Theorem 5.14, and Hypothesis 7.1,*

$$A_{\text{toy}} < B_{\zeta,2} \quad \implies \quad \rho \notin \{\zeta = 0\}.$$

Gap 8.4. Compose Theorem 5.14 (lower bound) with Hypothesis 7.1 (upper bound) at matched (m, s) ; show the exponent gap survives the polynomial whitening loss (Lemma 4.8) and the polynomial Gram lower bound (Lemma 3.6). The proof is short — arithmetic on absolute exponents — once the inputs are in place.

Remark 8.5 (Energy-supply route, if invoked). If Hypothesis 7.1 cannot be discharged from the local jet-Gram geometry alone for the two-point case, the source-energy lower bound Hypothesis 12.3 of Section 12, transferred via Lemma 12.5, supplies the missing polynomial-in- Q room. Theorem 8.3 is then conditional on that source as well as on the local inputs.

8.3. Four-point comparison in the symmetric quartet. The four-point comparison uses the full symmetry packet $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$. The four-center jet-Gram block carries strictly more rigidity than the two-point cross-block: the toy visibility lower bound holds in a refined quotient invisible to the two-point comparison, and the matching suppression hypothesis is correspondingly stronger. The four-point theorem is stated independently of Theorem 8.3.

Theorem 8.6 (Four-point rigidity). *Let ρ be a hypothetical off-line zero and $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$ its symmetry quartet. Under the four-point analogue of Theorem 5.14 and Hypothesis 7.3,*

$$A_{\text{toy}}^{(4)} < B_{\zeta,4} \quad \implies \quad \{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\} \not\subset \{\zeta = 0\}.$$

Gap 8.7. State and prove the four-center analogue $\widehat{\Omega}_{\zeta}^{(4)}$ of $\widehat{\Omega}_{\zeta}$; the four-point projection $\Pi_{\text{trans}}^{(4)}$; and the four-point analogue of Theorem 5.14 matching Hypothesis 7.3. Assemble the exponent inequality. v1 prose to mine: the local projective rigidity of the one-pair and multi-pair families section, together with the corrected fixed-target quotient package.

Remark 8.8 (Energy-supply route, if invoked). The four-point analogue of Remark 8.5 applies: if Hypothesis 7.3 cannot be proved from local geometry alone, Hypothesis 12.3 of Section 12 supplies the missing room via the four-point branch of Lemma 12.5, separating $A_{\text{toy}}^{(4)}$ from $B_{\zeta,4}$.

9. N -POINT ODD-MOMENT CANCELLATION

paper: [LIVE] v1: [migrated] referee: [in-progress] sympy: [verified] rh/sympy/sec-n-point-odd-moment-cancellation.py
sim: [verified] rh/simulation/sec-n-point-odd-moment-cancellation.py lean: [n/a]

The two-point scalar channel, and any future four-point analogue, may produce Taylor remainder terms in the separation variable. This section supplies only the algebraic cancellation package used to remove low-order odd terms: an explicit signed discrete N -point measure $\mu_{N,Q}$ on s -space whose low odd moments vanish by construction. It does not by itself prove that the remainders are below $Q^{-A_{\text{toy}}}$, below $Q^{-B_{\zeta}}$, or uniformly negligible in height; those estimates belong to the two-/four-point comparison and aggregation sections.

9.1. Coefficients and moment identities.

Definition 9.1 (N -point coefficients). For each integer $N \geq 1$ and $1 \leq j \leq N$,

$$(9.1) \quad a_j^{(N)} := (-1)^{j+1} \frac{j \binom{2N}{N+j}}{\binom{2N-2}{N-1}}.$$

Lemma 9.2 (Moment identities for $a_j^{(N)}$). *The coefficients $a_j^{(N)}$ of (9.1) satisfy*

$$(9.2) \quad \sum_{j=1}^N a_j^{(N)} = 1,$$

$$(9.3) \quad \sum_{j=1}^N a_j^{(N)} j^{2r+1} = 0 \quad (0 \leq r \leq N-2),$$

$$(9.4) \quad \sum_{j=1}^N |a_j^{(N)}| = 2N - 1.$$

Consequently, for every integer $k \geq 0$,

$$(9.5) \quad \left| \sum_{j=1}^N a_j^{(N)} j^k \right| \leq (2N-1) N^k.$$

Proof. Write $D_N = \binom{2N-2}{N-1}$, so $a_j^{(N)} = (-1)^{j+1} j \binom{2N}{N+j} / D_N$.

Normalization. The identity

$$j \binom{2N}{N+j} = N \left(\binom{2N-1}{N+j-1} - \binom{2N-1}{N+j} \right)$$

gives

$$\sum_{j=1}^N (-1)^{j+1} j \binom{2N}{N+j} = N T_N,$$

where

$$T_N := \sum_{j=1}^N (-1)^{j+1} \left(\binom{2N-1}{N+j-1} - \binom{2N-1}{N+j} \right).$$

Putting $k = N + j - 1$ in the first sum and $k = N + j$ in the second, and using $\binom{2N-1}{2N} = 0$, one obtains

$$T_N = 2 \sum_{k=N}^{2N-1} (-1)^{k-N} \binom{2N-1}{k} - \binom{2N-1}{N}.$$

The partial alternating-binomial identity gives

$$\sum_{k=N}^{2N-1} (-1)^{k-N} \binom{2N-1}{k} = D_N,$$

and $\binom{2N-1}{N} = ((2N-1)/N) D_N$. Hence $T_N = D_N/N$, and so $\sum_j a_j^{(N)} = D_N^{-1} N T_N = 1$.

Odd-moment vanishing. For any polynomial $P(k)$ of degree $< 2N$,

$$(9.6) \quad \sum_{k=0}^{2N} (-1)^k \binom{2N}{k} P(k) = 0,$$

because the operator on the left is the $2N$ -th finite difference of P at the origin and any polynomial of degree $< 2N$ is annihilated by it. Apply (9.6) to $P(k) = (k - N)^{2r+2}$ for $0 \leq r \leq N - 2$ (so $2r + 2 < 2N$). After the change of variables $j = k - N$, this becomes

$$\sum_{j=-N}^N (-1)^j \binom{2N}{N+j} j^{2r+2} = 0.$$

The summand is even in j : the power j^{2r+2} is even, $(-1)^{-j} = (-1)^j$, and $\binom{2N}{N-j} = \binom{2N}{N+j}$. The $j = 0$ term vanishes, so

$$2 \sum_{j=1}^N (-1)^j \binom{2N}{N+j} j^{2r+2} = 0.$$

Since

$$\sum_{j=1}^N a_j^{(N)} j^{2r+1} = -D_N^{-1} \sum_{j=1}^N (-1)^j \binom{2N}{N+j} j^{2r+2},$$

we obtain (9.3).

ℓ^1 control. The signs of $a_j^{(N)}$ alternate, so $\sum_{j=1}^N |a_j^{(N)}| = D_N^{-1} \sum_{j=1}^N j \binom{2N}{N+j}$. Telescoping the same identity used for normalization gives $\sum_{j=1}^N j \binom{2N}{N+j} = N \binom{2N-1}{N}$, and the standard identity $N \binom{2N-1}{N} = (2N-1) \binom{2N-2}{N-1}$ yields $\sum_{j=1}^N |a_j^{(N)}| = 2N-1$.

The bound (9.5) follows from $|\sum a_j j^k| \leq \sum |a_j| j^k \leq (2N-1) N^k$. $\square$

Remark 9.3 (The case $N = 1$). For $N = 1$, the range $0 \leq r \leq N - 2$ is empty, so there is no low odd moment to cancel. The normalization gives $a_1^{(1)} = 1$, and Corollary 9.9 says that the first odd jet is already the first derivative. Thus genuine cancellation starts at $N \geq 2$, while the formulas remain valid for $N = 1$.

9.2. The discrete N -point measure.

Definition 9.4 (N -point measure and exponential symbol). For each height parameter $Q > 0$, let $\mu_{N,Q}$ be the signed discrete measure on $\mathbb{R}$ supported on $\{j/Q^2 : 1 \leq j \leq N\}$ with masses $a_j^{(N)}$ of Definition 9.1. Its exponential symbol is

$$(9.7) \quad \widehat{\mu}_{N,Q}(z) := \sum_{j=1}^N a_j^{(N)} e^{jz/Q^2}.$$

Here z is a formal or complex local variable. No factor of i is included in this convention; the hat denotes the generating symbol used for moment cancellation, not a unitary Fourier transform.

Lemma 9.5 (Odd-jet vanishing of $\widehat{\mu}_{N,Q}$). For every $N \geq 1$ and every $Q > 0$,

$$(9.8) \quad \widehat{\mu}_{N,Q}(0) = 1,$$

and

$$(9.9) \quad \partial_z^{2r+1} \widehat{\mu}_{N,Q}(0) = 0 \quad (0 \leq r \leq N - 2).$$

Proof. Differentiating (9.7),

$$\partial_z^k \widehat{\mu}_{N,Q}(0) = Q^{-2k} \sum_{j=1}^N a_j^{(N)} j^k.$$

For $k = 0$ this is $\sum_j a_j^{(N)} = 1$ by (9.2). For $k = 2r + 1$ with $0 \leq r \leq N - 2$ the inner sum is zero by (9.3). $\square$

Lemma 9.6 (Convolution stability). Let $\widehat{\phi} : \mathbb{R} \rightarrow \mathbb{R}$ be even and C^∞ near 0, and let $\eta \in \mathbb{R}$. Then for every $N \geq 1$ and every $Q > 0$,

$$(9.10) \quad \partial_z^{2r+1} (\widehat{\phi}(\eta z) \widehat{\mu}_{N,Q}(z))|_{z=0} = 0 \quad (0 \leq r \leq N - 2).$$

Proof. By the Leibniz rule,

$$\partial_z^{2r+1} (\widehat{\phi}(\eta z) \widehat{\mu}_{N,Q}(z))|_{z=0} = \sum_{a+b=2r+1} \binom{2r+1}{a} \eta^a \widehat{\phi}^{(a)}(0) \partial_z^b \widehat{\mu}_{N,Q}(0).$$

With $a + b$ odd, exactly one of a, b is odd. If a is odd, $\widehat{\phi}^{(a)}(0) = 0$ since $\widehat{\phi}$ is even. If b is odd, then $b \leq 2r + 1 \leq 2N - 3$, so $\partial_z^b \widehat{\mu}_{N,Q}(0) = 0$ by Lemma 9.5. Every summand vanishes. $\square$

Remark 9.7 (Even bump hypothesis is necessary). The even-bump hypothesis in Lemma 9.6 is load-bearing: if $\widehat{\phi}$ carries an odd component, the odd part of the product no longer vanishes at any order below $2N - 1$.

9.3. Algebraic-difference identity.

Lemma 9.8 (Closed form for $A_N(t) - A_N(t^{-1})$). With $A_N(t) := \sum_{j=1}^N a_j^{(N)} t^j$,

$$(9.11) \quad A_N(t) - A_N(t^{-1}) = \frac{(-1)^N N}{\binom{2N-2}{N-1}} t^{-N} (1-t)^{2N-1} (1+t).$$

Proof. Set $F_N(t) := \sum_{j=-N}^N (-1)^j \binom{2N}{N+j} t^j$. By the binomial theorem, $F_N(t) = (-1)^N t^{-N} (1-t)^{2N} (1+t)^{2N}$. Differentiating once and multiplying by t ,

$$tF'_N(t) = -N(-1)^N t^{-N} (1-t)^{2N-1} (1+t).$$

On the other hand, the symmetry $(-1)^{-j} \binom{2N}{N-j} = (-1)^j \binom{2N}{N+j}$ gives $tF'_N(t) = \sum_{j=1}^N j(-1)^j \binom{2N}{N+j} (t^j - t^{-j})$. Multiplying by $-1/\binom{2N-2}{N-1}$ and re-introducing the $(-1)^{j+1}$ prefactor of (9.1) yields (9.11). $\square$

Corollary 9.9 (First surviving odd derivative). *For every $N \geq 1$ and every $Q > 0$,*

$$(9.12) \quad \partial_z^{2N-1} \hat{\mu}_{N,Q}(0) = \frac{(-1)^{N+1} (2N-1) N! (N-1)!}{Q^{2(2N-1)}}.$$

Proof. Substituting $t = e^{z/Q^2}$ in (9.11), the left-hand side equals $\hat{\mu}_{N,Q}(z) - \hat{\mu}_{N,Q}(-z)$. Near $z = 0$,

$$1 - e^{z/Q^2} = -\frac{z}{Q^2} + O(z^2), \quad e^{-Nz/Q^2} = 1 + O(z), \quad 1 + e^{z/Q^2} = 2 + O(z).$$

Therefore

$$\hat{\mu}_{N,Q}(z) - \hat{\mu}_{N,Q}(-z) = \frac{(-1)^N N}{\binom{2N-2}{N-1}} \left(-\frac{z}{Q^2}\right)^{2N-1} \cdot 2 + O(z^{2N}) = \frac{2(-1)^{N+1} N z^{2N-1}}{\binom{2N-2}{N-1} Q^{2(2N-1)}} + O(z^{2N}).$$

The $(2N-1)$ -fold derivative at $z = 0$ of the left-hand side is $2 \partial_z^{2N-1} \hat{\mu}_{N,Q}(0)$; on the right-hand side it extracts $(2N-1)!$ times the leading coefficient. Equating and using $\binom{2N-2}{N-1} = (2N-2)!/((N-1)!)^2$ together with $N(2N-1)!/(2N-2)! = N(2N-1)$ yields (9.12). $\square$

Remark 9.10 (Where the order- $2N-1$ vanishing comes from). The factor $(1-t)^{2N-1}$ in (9.11) is the algebraic source of the order- $2N-1$ odd-jet vanishing of $\hat{\mu}_{N,Q}$: substituting $t = e^{z/Q^2}$, $\hat{\mu}_{N,Q}(z) - \hat{\mu}_{N,Q}(-z)$ inherits a zero of order exactly $2N-1$ at $z = 0$, with leading coefficient given by Corollary 9.9.

Remark 9.11 (Scope of §9). This section supplies parity machinery only. No analytic comparison, finite-scale block estimate, source-energy transfer, or two-/four-point exponent estimate appears here. The downstream consumers of Lemma 9.2 and Lemma 9.5 must separately prove that the scalar quantity to which $\mu_{N,Q}$ is applied has a Taylor expansion with the required even-bump/parity structure and with remainders small enough for the relevant exponent comparison. In particular, this section does not prove the two-point or four-point suppression hypotheses; it only provides the finite algebra used by those later arguments.

10. AGGREGATION AND ZERO-FREE STRIP

paper: [LIVE] v1: [migrated] referee: [in-progress] sympy: [n/a] sim: [n/a] lean: [n/a]

The preceding sections separate local exclusion from global assembly. The local comparison theorems of Section 8 exclude admissible packets in the two-point scalar branch or in the four-point full-quartet branch. The branch-exhaustion input (Hypothesis 8.1) says that these two branches cover all admissible off-line packets produced by the branch-entry construction. The present section records the final high-height assembly. It does not prove branch entry, actual-zeta suppression, source-energy transfer, or the analytic estimates needed to consume the N -point cancellation of Section 9; those inputs must already be included in the rigidity/suppression theorems being invoked.

Hypothesis 10.1 (High-height branch entry). There exists a height $T_{\text{entry}} > 0$ such that every hypothetical off-line zero

$$\rho = \beta + i\gamma, \quad \beta \neq \frac{1}{2}, \quad |\gamma| \geq T_{\text{entry}},$$

produces an admissible off-line packet in the local jet–Gram framework, with all normalizations, retained subdomains, and loss ledgers required by Hypothesis 8.1 and by the applicable rigidity theorem.

Remark 10.2 (All-height alternative versus finite verification). The high-height formulation is a proof-management device, not an intrinsic feature of the desired theorem. There are two possible ways to remove the finite-height input in Section 11.

First, one may strengthen Hypothesis 10.1 and the local rigidity inputs so that they apply at every nontrivial-zero height, with no asymptotic lower-height cutoff. This would require replacing the large- T estimates used in the local chart, whitening, toy high-frequency limit, and packet-retention arguments by uniform finite-height estimates. In that case one may take $T_{\text{entry}} = 0$ and no numerical finite-height verification is needed for the nontrivial zeros.

Second, one may keep the high-height proof but make the resulting threshold T_0 effective. If a certified computation verifies that all zeros with $0 < |\gamma| \leq H_{\text{cert}}$ lie on the critical line and if $T_0 \leq H_{\text{cert}}$, then the finite-height verification input of Hypothesis 11.1 follows from that certificate. For example, Platt–Trudgian verify the Riemann hypothesis rigorously up to height $3 \cdot 10^{12}$ by interval arithmetic. If T_0 is ineffective, or if $T_0 > H_{\text{cert}}$, then a separate finite-height certificate up to T_0 is still required.

Remark 10.3 (Role of the N -point section in the final assembly). Section 9 supplies algebraic scalar moment cancellation. If a two-point or four-point suppression theorem uses that cancellation, then the corresponding analytic remainder estimates, polynomial-weight density losses, and exponent bookkeeping must be absorbed in that suppression or rigidity theorem before Theorem 10.4 is invoked. The final assembly below does not itself turn the N -point algebra into an analytic suppression estimate.

Theorem 10.4 (Zero-free strip above an absolute height). *Assume Hypothesis 10.1, Hypothesis 8.1, and the inputs of Theorems 8.3 and 8.6 with absolute high-height thresholds. Then there exists an absolute constant $T_0 > 0$ such that*

$$(10.1) \quad \zeta(\beta + i\gamma) \neq 0 \quad \text{for every } \beta \neq \tfrac{1}{2} \text{ and } |\gamma| \geq T_0.$$

Proof. Let $T_0^{(2)}$ and $T_0^{(4)}$ be the high-height thresholds for the two-point and four-point rigidity theorems, respectively, and let

$$T_0 := \max\{T_{\text{entry}}, T_0^{(2)}, T_0^{(4)}\}.$$

Suppose, for contradiction, that $\rho = \beta + i\gamma$ is a nontrivial zero of ζ with $\beta \neq \frac{1}{2}$ and $|\gamma| \geq T_0$. By Hypothesis 10.1, ρ produces an admissible off-line packet. By Hypothesis 8.1, this packet is excluded by either the two-point scalar branch of Theorem 8.3 or the four-point full-quartet branch of Theorem 8.6.

If the two-point branch applies, then $|\gamma| \geq T_0^{(2)}$, and Theorem 8.3, under its stated inputs, contradicts the existence of ρ . If the four-point branch applies, then $|\gamma| \geq T_0^{(4)}$, and Theorem 8.6, under its stated inputs, contradicts the existence of the symmetry quartet generated by ρ . Both alternatives are impossible. Hence (10.1) holds. $\square$

Remark 10.5 (No additional zero-counting loss). Once Hypothesis 10.1, Hypothesis 8.1, and the relevant local rigidity theorem are available with absolute thresholds, the final high-height assembly does not accumulate a loss over the set of zeros. Each hypothetical off-line zero above T_0 is contradicted individually. Any polynomial-weight density loss, N -point moment-cancellation loss, or source-energy loss must already have been charged in the proof of the suppression or rigidity input being used.

Remark 10.6 (Scope of §10). This section proves only the formal high-height implication from branch entry, branch exhaustion, and local rigidity to a zero-free strip. It does not address heights $|\gamma| < T_0$, and it does not make T_0 effective. The finite height range is a separate input in Section 11 unless all constants have been made explicit and lie below an already certified verification height.

11. THE RIEMANN HYPOTHESIS

Theorem 10.4 excludes off-line zeros above the height T_0 supplied by the high-height assembly. To conclude the full Riemann hypothesis, the bounded interval $|\gamma| < T_0$ must also be certified. If T_0 is not effective, this finite-height certification remains a separate input.

Hypothesis 11.1 (Finite-height verification). For the height T_0 appearing in Theorem 10.4, every nontrivial zero $\rho = \beta + i\gamma$ of ζ with $|\gamma| < T_0$ satisfies $\beta = \frac{1}{2}$.

Theorem 11.2 (Conditional Riemann hypothesis). *Assume the inputs of Theorem 10.4 and Hypothesis 11.1. Then every nontrivial zero of ζ satisfies $\operatorname{Re} s = \frac{1}{2}$.*

Proof. Let $\rho = \beta + i\gamma$ be a nontrivial zero of ζ . If $|\gamma| \geq T_0$, then $\beta = \frac{1}{2}$ by Theorem 10.4. If $|\gamma| < T_0$, then $\beta = \frac{1}{2}$ by Hypothesis 11.1. Thus every nontrivial zero lies on the critical line. $\square$

Remark 11.3 (Effectivity and numerical verification). Certified finite-height verification may be used once the height threshold is made explicit. Let H_{cert} denote a rigorous certified height up to which every nontrivial zero has been verified to lie on the critical line. If the proof produces an explicit $T_0 \leq H_{\text{cert}}$, then Hypothesis 11.1 follows from that certificate. Known computations include the rigorous interval-arithmetic verification of Platt–Trudgian up to $H_{\text{cert}} = 3 \cdot 10^{12}$. If the proof only produces a non-effective T_0 , then known finite computations cannot be invoked automatically; one must either make T_0 effective and compare it with a certified height, prove all-height applicability of the local argument, or supply a finite verification certificate for the particular T_0 .

12. SOURCE-ENERGY SUPPLY (OPTIONAL)

paper: [LIVE] v1: [not-started] referee: [not-started] sympy: [not-started] sim: [not-started] lean: [not-started]

This section is self-contained: its sole role is to supply a single source-energy lower bound that Theorems 8.3 and 8.6 may invoke when their standalone forms in Section 8 are insufficient. Nothing earlier in the paper depends on this section.

12.1. Canonical source curvature.

Definition 12.1 (Canonical source curvature). The canonical source curvature of ζ on the chart of Definition 2.4 is

$$(12.1) \quad q''_{\text{can}} := \partial_t^3 Z(t),$$

where $Z(t)$ is the real critical-line scalar of ζ in the same chart.

Gap 12.2. Identification of q''_{can} with $\partial_t^3 Z$ is a convention; chart-regularity and zero-separation hypotheses of Definition 2.4 carry over verbatim. v1 prose to mine: the upstream Gate-1 arithmetic inputs section.

12.2. Source-energy bound.

Hypothesis 12.3 (Source-energy lower bound). On retained packet intervals $I \subset I_T$, the canonical source curvature satisfies

$$(12.2) \quad \int_I |q''_{\text{can}}(m)|^2 dm \geq Q^{-A_E} |I|$$

on a positive polynomial-weight subfamily, for an absolute exponent A_E .

Gap 12.4. polynomial lower bound on $\partial_t^3 Z$ across the retained packet family. This is an analytic estimate on ζ and not on the local jet-Gram geometry; it is what makes Section 12 optional. v1 prose to mine: the source-energy mass-good packet block.

12.3. Energy-to-suppression transfer.

Lemma 12.5 (Source energy to transverse suppression). *Hypothesis 12.3, after carrier decomposition and packet trace normalization, implies one of Hypothesis 7.1 (with exponent $B_{\zeta,2} = B_{\zeta,2}(A_E) > A_{\text{toy}}$) or Hypothesis 7.3 (with exponent $B_{\zeta,4} = B_{\zeta,4}(A_E) > A_{\text{toy}}^{(4)}$), according to which carrier transfer is established into the corresponding transverse channel of $\widehat{\Omega}_{\zeta}(s; m)$ or $\widehat{\Omega}_{\zeta}^{(4)}$.*

Gap 12.6. Carrier decomposition and packet trace step taking $\int |q''_{\text{can}}|^2$ on the source side into a lower bound on $\Pi_{\text{trans}} \widehat{\Omega}_{\zeta}$ (two-point case) or $\Pi_{\text{trans}}^{(4)} \widehat{\Omega}_{\zeta}^{(4)}$ (four-point case). v1 prose to mine: the certified downstream status after the canonical source-graded closure section, and the carrier-corrected source bridge package.

APPENDIX A. ARCHIVE MAP

paper: [DIAGNOSTIC] v1: [n/a] referee: [n/a] sympy: [n/a] sim: [n/a] lean: [n/a]

The previous large draft is preserved in place at `rh/proof_of_rh.tex` and is treated as a quarry from which prose and computations may be migrated, after restatement and refereeing, into the present file. Pointers to relevant v1 sections are given inline at each `wip` mark above; the table below summarizes the principal mining correspondences.

Rebuild section	v1 source (in <code>proof_of_rh.tex</code>)
§2	Local kernel and jet normalization.
§3	Finite- s local blocks; jet-limit local blocks.
§4	Corrected finite- s holomorphic whitening.
§5	Toy anomaly and transverse obstruction.
§6	Leading value matrix and canonical calibration.
§7	Odd holomorphic transverse channel; corrected fixed-target quotient package.
§8	Local projective rigidity of the one-pair and multi-pair families; finite coefficient table contracts.
§9	N -point odd-moment cancellation.
§10, §11	Master conditional theorem; introduction.
§12	Source-energy mass-good packet block; carrier-corrected source bridge package; upstream Gate-1 arithmetic inputs.

The v1 sections “Endgame architecture and remaining propagation problem outside the clean mixed four-point case,” “Downstream local closure package: Gate 2, wall escape, and carrier rows,” “Unified finite-survival architecture,” “Current proof state after the local package audits,” “How the older framing maps to the new framing,” and “No-go and audit ledger” are deliberately not migrated; their content is route history and process bookkeeping rather than active proof material.

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